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Explicit methods for stress wave propagation

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Project Brief

A Hopkinson bar is a high speed transient load measurement device. It is, essentially, a long narrow bar to which a dynamic load is applied at one end. The load generates stress waves that travel along the bar and are captured by strain gauges, from which the load history is extracted using stress wave theory. However, the current useful range of the Hopkinson bar technique is limited by the incomplete understanding of higher order stress wave propagation phenomena. This project forms part of a series aimed at ultimately producing a 2D axi-symmetric finite element analysis (FEA) code in python to analyse higher order wave propagation in slender bars. The codes will be open-source and accessible, i.e. based on theory or techniques that undergraduate engineering students can master within a final year project. Previous projects have resulted in a code that can determine the vibration modes and frequencies of slender bar. The aim of this project is to critically evaluate the current code and modify it for explicit FEA, i.e. a numerical temporal integration scheme that does not require matrix inversion. The standard Central Differences scheme must be implemented. Other potential schemes, such as multi-stepping can also be attempted. This work will include an in-depth study of stress wave theory, vibration of bars and FEA. Note: While this project is not a group project, it is one of several projects focusing on 2D FEA and students with these projects are encouraged to work together.

Supervisor:	T. Cloete
Word Count:	6942

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Abstract

Hopkinson pressure bar measurements infer transient loads or material behaviour from strains recorded on the bar surface. Because the bar has a finite radius, the stress pulse disperses as it travels, so the measured signal differs from the one at the loaded end. Dispersion corrections based on Pochhammer–Chree theory account for this, but only for the first mode. Finite element models can capture the full wave behaviour, but only once the code itself is trusted. This project developed and verified an explicit finite element solver for elastic stress wave propagation in bars, as a base for later work on higher-mode dispersion. The solver is written in Python. It uses axisymmetric Q4 elements, offers consistent, lumped and combined mass matrices, and integrates in time with the central difference method, with the critical time step found from the largest eigenvalue of the system. In the verification tests, the element passed the patch test and reproduced the consistent nodal loads to machine precision. Energy and momentum were conserved to round-off, and the strain error fell at second order or faster as the axial mesh was refined, with consistent mass about nine times less accurate than lumped mass on refined meshes. A free-end reflection test confirmed the wave speed and reflection, and a square-pulse test showed the expected dispersion of each mass matrix, with the consistent matrix producing non-physical precursors ahead of the wavefront. The solver was then validated against first-mode Pochhammer–Chree theory following Govender et al. (2006). Above about 15 elements per wavelength, the phase velocity was within 1 % of theory, and the strain–time history at a gauge lagged the first-mode prediction by only 0.08 µs. Lumped mass is recommended, being the most accurate on refined meshes while allowing the cheapest steps and the largest time step. Mesh resolution is currently limited by dense matrix storage, so moving to sparse storage is the first step towards resolving higher modes.

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I would like to thank my supervisor, Dr Trevor Cloete, for his guidance throughout this project. He encouraged independent thinking and gave me the freedom to explore my own ideas, while keeping me focused on what mattered most. I am also grateful for his course notes, on which much of the formulation is based. Finally, I thank my family and friends for their support during my final year.

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Glossary of Terms and Acronyms

Symbols

Symbol	Definition	Unit
a	Bar radius	mm
c_0	Bar wave speed, $\sqrt{E/\rho}$	mm/s
$c_g,\ c_p$	Group and phase velocity	mm/s
E	Young's modulus	MPa
f	Frequency	Hz
L	Bar length	mm
N_i	Shape function of node i	–
$n_r,\ n_z$	Number of radial and axial elements	–
P	Axial momentum	N s
$q(t)$	Applied axial traction, positive in compression	MPa
$r,\ \theta,\ z$	Radial, circumferential and axial coordinates	mm, rad, mm
$T,\ U$	Kinetic and strain energy	N mm
t	Time	s
$u_r,\ u_z$	Radial and axial displacement	mm
W	External work	N mm
z_g	Axial position of the strain gauge	mm
β	Weighting factor of the combined mass matrix	–
γ_{rz}	Shear strain in the r–z plane	–
$\Delta r,\ \Delta z$	Radial and axial element size	mm
$\Delta t,\ \Delta t_{cr}$	Time step and critical time step	s
$\varepsilon_r,\ \varepsilon_\theta,\ \varepsilon_z$	Radial, hoop and axial normal strain	–
λ	Wavelength	mm
ν	Poisson's ratio	–
$\xi,\ \eta$	Natural (element) coordinates	–
ρ	Density	tonne/mm^3
$\sigma_r,\ \sigma_\theta,\ \sigma_z$	Radial, hoop and axial normal stress	MPa
τ_{rz}	Shear stress in the r–z plane	MPa
$\omega,\ \omega_{max}$	Angular frequency; highest natural frequency of the mesh	rad/s
$[B]$	Strain–displacement matrix	mm^{-1}
$[E]$	Elasticity matrix	MPa
$[J]$	Jacobian matrix	mm
$\lvert J\rvert,\ J$	Determinant of the Jacobian matrix	mm^2

$[K]$, $[k]$	Global and element stiffness matrix	N/mm
$[M]$	Global mass matrix (consistent, lumped or combined)	tonne
$[N]$	Shape function matrix	–
$\{d\}$	Nodal displacement vector	mm
$\{\dot{d}\}$, $\{\ddot{d}\}$	Nodal velocity and acceleration vectors	mm/s, mm/s^2
$\{F\}$	Global nodal force vector	N

Abbreviations

Abbreviation	Meaning
CFL	Courant–Friedrichs–Lewy
DOF	Degree of freedom
FEA	Finite element analysis
GWS	Gaussian-windowed sine
HPB	Hopkinson pressure bar
HRZ	Hinton–Rock–Zienkiewicz (mass lumping)
Q4	Four-node quadrilateral element

Terms

Term	Meaning
Cutoff frequency	Frequency below which a higher mode cannot propagate
Dispersion	Dependence of wave speed on frequency, so a pulse changes shape as it travels
Group velocity	Speed at which the energy of a pulse travels
Isoparametric element	Element whose geometry and displacements use the same shape functions
Patch test	Check that elements reproduce a constant-strain state exactly, whatever their shape
Phase velocity	Speed of a single frequency component of a wave
Pochhammer–Chree theory	Exact solution for longitudinal waves in an infinite elastic cylinder
Precursor	Non-physical signal arriving ahead of the true wavefront in a discretised model
Traction	Force per unit area on a surface

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1

Introduction

1.1. Background and Motivation

As outlined in the project brief, the useful range of the Hopkinson pressure bar (HPB) is limited by dispersion: in a bar of finite radius, waves of different frequency travel at different speeds, and higher-order modes appear at high frequencies, so the pulse measured at the strain gauge no longer matches the pulse at the specimen (Rigby et al., 2018). Commercial packages such as Abaqus and LS-DYNA can model this behaviour, but they are closed-source, so their formulations cannot be inspected or modified, and licences are often unavailable to undergraduate students. This series of projects therefore aims to build an open finite element analysis (FEA) code in which every component is derived from first principles and independently verified, so that it can be used to investigate dispersion and validated against analytical solutions such as Pochhammer–Chree theory.

1.2. Aims and Objectives

The previous project in this series (Groenewald, 2024) developed an axisymmetric code for the natural frequencies and mode shapes of slender bars, but could not model transient wave propagation. The aim of this project is to develop an explicit solver using the central difference scheme, verify it through targeted tests and validate it against Pochhammer–Chree theory. The objectives are to:

1. Critically evaluate the existing code from the previous project
2. Derive the formulation for explicit analysis with axisymmetric four-node quadrilateral (Q4) elements, including the stiffness matrix, the lumped, consistent and combined mass matrices, the consistent nodal loads, the central difference scheme and its critical time step
3. Implement the solver in Python
4. Verify the solver through a patch test, a consistent nodal load check, energy and momentum balance, mesh convergence, free-end reflection and a square-pulse test
5. Compare the accuracy of the three mass matrices

6. As an extension beyond the brief, implement the Pochhammer–Chree frequency equation and validate the solver against it and against the results of Govender et al. (2006)

1.3. Scope

The work is limited to a slender cylindrical bar that is linear elastic, isotropic and homogeneous, with small strains and no damping. The geometry and loading are axisymmetric, so the problem reduces to two dimensions, and non-axisymmetric modes such as bending and torsion are excluded. Only the central difference scheme is implemented; other time integration schemes, such as Adams–Bashforth or implicit Newmark methods, are outside the scope. The Pochhammer–Chree frequency equation is solved numerically by root-finding, without a full derivation of its Bessel-function solution, and validation is limited to the regime in which only the first mode propagates.

2

Literature Review

2.1. The Hopkinson Bar and Dispersion

The Hopkinson pressure bar (HPB) is widely used to measure transient loads from impact and blast, as it combines the durability needed to withstand high pressures with the sensitivity needed to resolve fast-changing signals (Rigby et al., 2018). A load applied at one end of a long, slender bar generates a stress wave that is recorded by strain gauges some distance away, and the load history is recovered by assuming that the wave arrives unchanged. Davies (1948) showed that this assumption breaks down at high frequencies: components of different frequency travel at different speeds, so the pulse changes shape as it propagates, and the strain at the bar surface no longer represents the average across the section. This dispersion distorts the high-frequency features that blast and impact measurements aim to capture, and limits the useful range of the technique (Rigby et al., 2018).

2.2. Pochhammer–Chree Theory

The exact description of longitudinal waves in an infinitely long, traction-free elastic cylinder was derived independently by Pochhammer and Chree (Graff, 1975, ch. 8). It relates the phase velocity of a sinusoidal wave to its wavelength, and Bancroft (1941) later evaluated it for the first mode across a range of Poisson's ratios. For long wavelengths, the first mode travels at the one-dimensional bar speed c_0; as the wavelength shortens, lateral inertia slows it towards the Rayleigh wave speed. The equation also has infinitely many higher modes, each of which propagates only above its own cutoff frequency. Individual crests travel at the phase velocity, while the energy of a pulse travels at the group velocity, which in the first mode is always below c_0 (Barr et al., 2020). Davies (1948) showed how stress and strain vary across the radius in each mode.

The theory therefore gives the speed of every mode, but not how much of a given signal travels in each, and it assumes a bar of infinite length. In this project it serves as the reference solution for validating the finite element results.

2.3. Dispersion Correction

The first practical dispersion corrections were developed by Gorham (1983) and Follansbee and Frantz (1983). A recorded signal is transformed into the frequency domain, the phase of each component is shifted according to its first-mode phase velocity, and the signal is transformed back, giving its shape at another point on the bar. Tyas and Watson (2001) added an amplitude correction for the radial variation of strain, so that a surface measurement can be related to the mean stress. Both the phase and the amplitude corrections assume that only the first mode is present.

Above the second-mode cutoff, each frequency is shared between two or more modes, in proportions that Pochhammer–Chree theory does not provide. Lee et al. (1995) found evidence of these higher modes experimentally. An alternative approach, which assigns each frequency entirely to the mode with the highest group velocity, was shown by Husemeyer (2011) to neglect a significant share of the signal. Barr et al. (2020) used a validated finite element model to measure how stress divides between the first four modes, sending windowed single-frequency pulses down a long bar until each mode separated in time, and showed that including the higher modes noticeably improved the correction of an experimental blast signal. The strain–time prediction used in this project is the first-mode phase correction of Gorham (1983) and Follansbee and Frantz (1983), and is therefore valid only below the second-mode cutoff (Section 3.10.3).

2.4. Finite Element Modelling of Dispersion

Since a finite element model of a bar solves the same equations of linear elastodynamics from which the Pochhammer–Chree solution is derived, dispersion and higher modes should emerge from it naturally (Barr et al., 2020). Govender et al. (2006) confirmed this for the first mode using an eigen-analysis in Abaqus, finding close agreement below 150 kHz and an error exceeding 1 % once the wavelength fell below about 17 axial element lengths, with accuracy depending strongly on the axial mesh density but only weakly on the radial density. Their explicit simulations agreed well with the first-mode correction below 100 kHz, but at 200 kHz the pulse split into two parts, one of which matched only a prediction including the second mode. Barr et al. (2020) later validated an LS-DYNA model in both regimes, measuring the first-mode phase velocity from the phase difference between two stations, an approach used experimentally by Bacon (1998) in viscoelastic bars.

Numerical effects must also be controlled. Both lumped and consistent mass matrices introduce numerical artifacts (Goudreau and Taylor, 1972), whose character depends on the matrix used (Cook et al., 2001, ch. 11), and the oscillations produced by central difference time integration can be difficult to distinguish from physical dispersion (Barr et al., 2020). Multi-mode studies also demand long, finely meshed bars: Barr et al. (2020) used a 10 m bar with 0.4 mm elements for their higher-mode analysis. These findings set the mesh targets and validation approach of this project, and motivate its comparison of mass matrices.

2.5. Previous Work in This Series

This project is part of a series of final-year projects aimed at an open, accessible finite element code for studying higher-mode propagation in Hopkinson bars. The only preceding project, by Groenewald (2024), developed an axisymmetric Python code for the static and modal analysis of slender bars, documented the axisymmetric formulation in detail, and verified the stiffness matrix with an axial patch test, a thick-walled cylinder and a rotating disc. Several issues remained. The modal frequencies matched analytical values only when the mass was doubled. Inspection of the code for this project showed that the lumped mass was in fact correct: the global stiffness matrix was unintentionally added to itself before the eigenvalue solution, and the doubled mass cancelled this error rather than correcting it. Incorrect mode shapes were also reported as the mesh was refined. The patch test used an undistorted mesh, only lumped mass schemes restricted to rectangular elements were implemented, and the element routines were copied into each script rather than shared. Inputs such as the nodal forces and prescribed displacements were entered as fixed arrays rather than generated from the mesh, so changing the mesh or loading required editing them by hand. Explicit time integration, although an objective, was not reached. Since an explicit solver depends directly on a correct mass matrix and on easily refined meshes, this code was judged unsuitable as a starting point for the present work.

2.6. Summary and Scope

Pochhammer–Chree theory provides the speed of each mode, and finite element models have been shown to reproduce it, but no open, verified explicit axisymmetric code yet exists for this series. This project develops such a code and validates it against Pochhammer–Chree theory and Govender et al. (2006) in the first-mode regime, as the necessary foundation for multi-mode studies such as that of Barr et al. (2020).

3

Methodology

For the reasons given in Section 2.5, a new modular solver was written rather than extending the earlier code, keeping only its use of the generalised eigenvalue problem to find the critical time step. This chapter describes the model, the element formulation, the loading, the time integration and its stability limit, and the implementation, followed by the verification and validation tests whose results are reported in Chapter 4.

3.1. Model and Assumptions

The bar is treated as a homogeneous, isotropic, linear elastic solid under small strains, with no damping. This suits an HPB, which is designed to remain elastic, and keeps the stiffness and mass matrices constant throughout the simulation.

3.1.1. Geometry and Boundary Conditions

The bar is a solid cylinder of length L and radius a (Table 3.2). Since the geometry and loading are symmetric about the axis, the problem reduces to the r–z half-section in Figure 3.1, with u_r and u_z as the only unknowns, so bending and torsional modes are not captured.

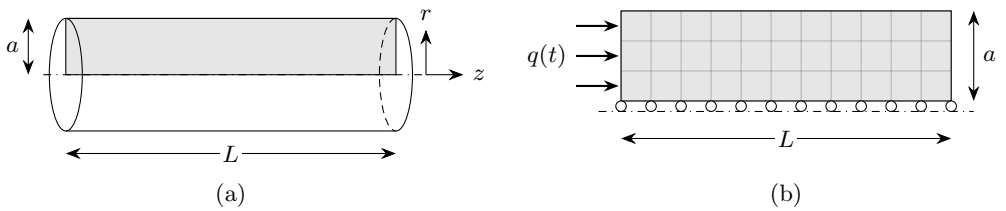


Figure 3.1: (a) The bar and the r–z half-section that is modelled. (b) The axisymmetric model, loaded by a uniform traction $q(t)$ on $z = 0$ with $u_r = 0$ on the axis; all other surfaces are traction-free (not to scale).

The boundary conditions are

$$\begin{aligned} \sigma_z = -q(t), \quad \tau_{rz} = 0 \qquad & \text{at } z = 0 \\ \sigma_r = 0, \quad \tau_{rz} = 0 \qquad & \text{at } r = a \\ \sigma_z = 0, \quad \tau_{rz} = 0 \qquad & \text{at } z = L \\ u_r = 0 \qquad & \text{at } r = 0 \end{aligned} \tag{3.1}$$

where $q(t)$ is the applied axial traction, positive in compression (Section 3.4). The lateral surface and far end are traction-free, so waves reaching the far end are reflected. Symmetry requires $u_r = 0$ on the axis, the only essential boundary condition. As in a real HPB, the bar is not restrained axially; its motion is resisted by inertia.

3.1.2. Material and Units

The code uses the consistent mm–N–tonne–s system, with stresses in MPa and density in tonne/mm^3. A standard steel is used for verification and the steel of Govender et al. (2006) for validation, so that the results can be compared directly with theirs (Table 3.1).

Table 3.1: Material properties. The square-pulse test uses $\nu = 0$ (Section 3.9.6).

Property	Symbol	Verification	Validation
Young's modulus	E	200 GPa	207 GPa
Poisson's ratio	ν	0.30	0.26
Density	ρ	7850 kg/m^3	7830 kg/m^3

3.2. Element Stiffness Matrix

The element stiffness matrix follows from the principle of minimum potential energy (Cloete, 2020; Cook et al., 2001). Writing the strains in terms of the nodal displacements, $\{\varepsilon\} = [B]\{d\}$, gives

$$[k] = \int_{A_e} [B]^T [E][B]\, 2\pi r \, dr \, dz, \tag{3.2}$$

where A_e is the element area in the r–z plane. The full derivation is given in Appendix A.

3.2.1. Isoparametric Interpolation

Each four-node element is mapped onto a square in natural coordinates (ξ, η), both running from -1 to 1 (Figure 3.2), and the displacements and geometry are interpolated with the same bilinear shape functions:

$$\begin{aligned} N_1 &= \tfrac{1}{4}(1-\xi)(1-\eta), \qquad N_2 = \tfrac{1}{4}(1+\xi)(1-\eta), \\ N_3 &= \tfrac{1}{4}(1+\xi)(1+\eta), \qquad N_4 = \tfrac{1}{4}(1-\xi)(1+\eta), \end{aligned} \tag{3.3}$$

$$\begin{Bmatrix} u_r \\ u_z \end{Bmatrix} = [N]\{d\}, \qquad r = \sum_{i=1}^{4} N_i r_i, \quad z = \sum_{i=1}^{4} N_i z_i. \tag{3.4}$$

Since the strains need derivatives in r and z, the two coordinate systems are linked through the Jacobian matrix, whose determinant also scales area:

$$\begin{Bmatrix} \partial/\partial\xi \\ \partial/\partial\eta \end{Bmatrix} = [J] \begin{Bmatrix} \partial/\partial r \\ \partial/\partial z \end{Bmatrix}, \qquad dr\,dz = |J|\,d\xi\,d\eta, \quad |J| = \det[J]. \tag{3.5}$$

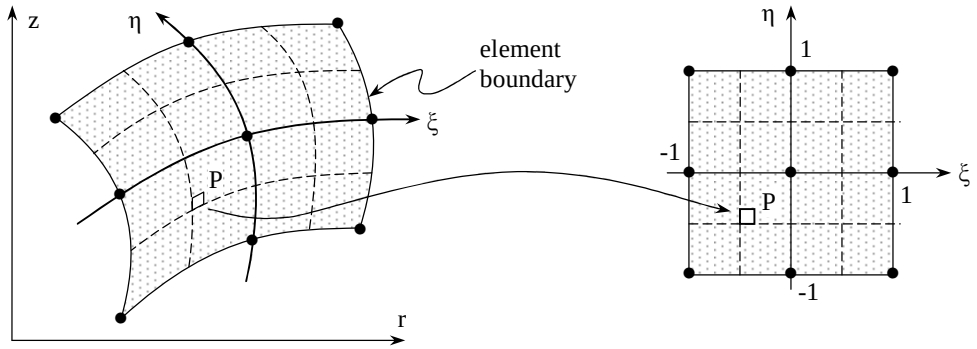


Figure 3.2: Mapping from natural coordinates to the physical element, adapted from Cloete (2020).

3.2.2. Strain–Displacement Matrix

Strains are derivatives with respect to r and z, but the shape functions are written in ξ and η, so $[B]$ is built in three steps: $[\partial N]$ differentiates the shape functions in ξ and η, $[\Gamma]$ converts these to r and z derivatives using $[J]^{-1}$, and $[H]$ combines them into the in-plane strains:

$$\begin{Bmatrix} \varepsilon_r \\ \varepsilon_z \\ \gamma_{rz} \end{Bmatrix} = [B_{2D}]\{d\}, \qquad [B_{2D}] = [H][\Gamma][\partial N]. \tag{3.6}$$

The axisymmetric element adds the hoop strain: a radial displacement stretches the ring of material at radius r from a circumference of $2\pi r$ to $2\pi(r + u_r)$,

$$\varepsilon_\theta = \frac{2\pi(r + u_r) - 2\pi r}{2\pi r} = \frac{u_r}{r}. \tag{3.7}$$

Since this depends on the displacement rather than its gradient, it is added directly as a fourth row of $[B]$:

$$[B] = \begin{bmatrix} [B_{2D}] \\ [B_\theta] \end{bmatrix}, \qquad [B_\theta] = \begin{bmatrix} \frac{N_1}{r} & 0 & \frac{N_2}{r} & 0 & \frac{N_3}{r} & 0 & \frac{N_4}{r} & 0 \end{bmatrix}. \tag{3.8}$$

This hoop row is the only part of $[B]$ that differs from a plane element, which is why the patch test is designed to check it.

3.2.3. Elasticity Matrix and Integration

With nothing varying in θ, the shear strains involving θ are zero, so $[E]$ keeps four components of Hooke's law, ordered as in $[B]$:

$$[E] = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 & \nu \\ \nu & 1-\nu & 0 & \nu \\ 0 & 0 & \frac{1-2\nu}{2} & 0 \\ \nu & \nu & 0 & 1-\nu \end{bmatrix}. \tag{3.9}$$

The $1/r$ terms make the integrand non-polynomial, so Equation (3.2) is evaluated with 3×3 Gauss quadrature, which integrates them more accurately than a 2×2 rule:

$$[k] \approx \sum_{i=1}^{3} \sum_{j=1}^{3} [B]^T [E][B]\, 2\pi r\, |J|\, w_i w_j. \tag{3.10}$$

Since the problem is linear, $[k]$ is computed once before time stepping, so the extra cost is negligible.

3.3. Mass Matrix

In the explicit scheme $[M]^{-1}$ is applied at every time step, so the way mass is distributed over the nodes affects both cost and accuracy. Three forms were implemented following Cook et al. (2001, ch. 11), with the derivations given in Appendix E. The consistent mass matrix follows from the kinetic energy using the same shape functions as the stiffness matrix,

$$[m_e] = \int_{A_e} \rho\, [N]^T [N]\, 2\pi r\, dr\, dz, \tag{3.11}$$

and is integrated exactly by the same 3×3 rule. It couples neighbouring nodes, so its inverse is dense. HRZ (Hinton–Rock–Zienkiewicz) lumping keeps only the diagonal of $[m_e]$ and scales it so that each direction carries the full element mass ρV_e, giving a diagonal $[M_{lump}]$ whose inverse is found term by term. In the spatial discretisation, consistent mass makes high-frequency waves travel too fast and lumped mass too slowly, so the combined matrix averages the two to let these errors partly cancel:

$$[M_{comb}] = \beta\, [M_{cons}] + (1-\beta)\, [M_{lump}], \qquad \beta = 0.5. \tag{3.12}$$

3.4. Loading

Dividing the end force equally between the face nodes would be wrong on an axisymmetric face, because the ring of material served by each node grows with radius. Instead, consistent nodal loads are used, which do the same work as the distributed traction (Cloete, 2020). For an element edge on $z = 0$ between radii r_1 and r_2, the axial nodal forces are

$$\{Q\} = \frac{2\pi q\, (r_2 - r_1)}{6} \begin{Bmatrix} 2r_1 + r_2 \\ r_1 + 2r_2 \end{Bmatrix} \tag{3.13}$$

(Appendix B). The two forces sum to the traction times the ring area, $\pi q(r_2^2 - r_1^2)$, and the assembled face loads to $\pi a^2 q$. Since they depend only on the face geometry, the load vector at each step is the unit-traction vector scaled by $q(t)$.

3.4.1. Input Pulses

Three pulses were used, each chosen for its frequency content (Figure 3.3; parameters in Table 3.2). The Gaussian-windowed sine (GWS) concentrates its energy near the carrier frequency, matching Govender et al. (2006), and is used for the energy and convergence tests. The Hann-windowed cosine falls to exactly zero at both ends, giving a clean pulse for the reflection test. The square pulse contains frequencies far beyond what the mesh can resolve, showing how each mass matrix handles them. All amplitudes are below yield.

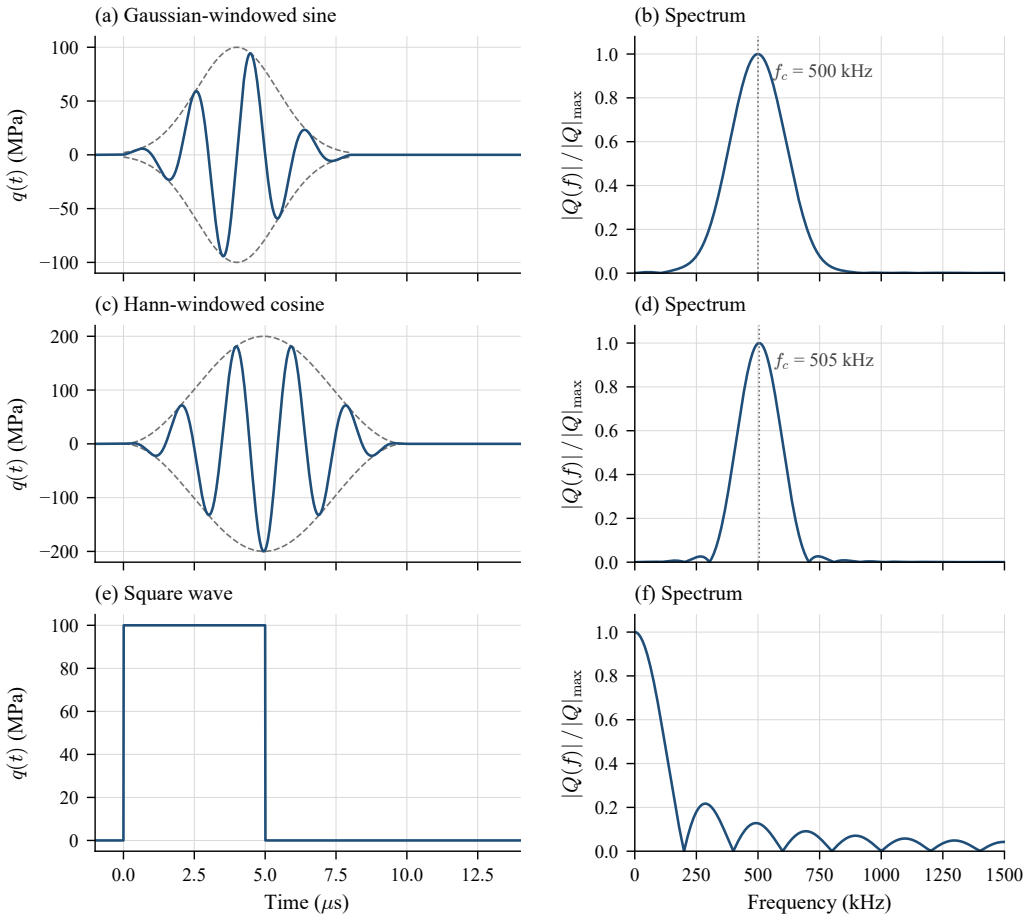


Figure 3.3: Applied traction pulses (left) and their normalised amplitude spectra (right): (a, b) the Gaussian-windowed sine, (c, d) the Hann-windowed cosine and (e, f) the square wave. The windowed pulses concentrate their energy around the carrier frequency f_c, while the square wave contains energy at all frequencies.

3.5. Explicit Time Integration

The assembled equations of motion,

$$[M]\{\ddot{d}\} + [K]\{d\} = \{F(t)\}, \tag{3.14}$$

are stepped forward with the central difference method (Cook et al., 2001, ch. 11), which approximates the velocity and acceleration at step n from the displacements on either side:

$$\{\dot{d}\}_n = \frac{\{d\}_{n+1} - \{d\}_{n-1}}{2\Delta t}, \qquad \{\ddot{d}\}_n = \frac{\{d\}_{n+1} - 2\{d\}_n + \{d\}_{n-1}}{\Delta t^2}. \tag{3.15}$$

Substituting the acceleration into Equation (3.14) gives

$$\{d\}_{n+1} = 2\{d\}_n - \{d\}_{n-1} + \Delta t^2 [M]^{-1} \left(\{F\}_n - [K]\{d\}_n\right). \tag{3.16}$$

Everything on the right is known, so no system of equations is solved at each step, which makes the method explicit. The axis radial displacements are removed from the system, enforcing $u_r = 0$ exactly. Since the bar starts at rest, a Taylor expansion about $t = 0$ gives the starting value

$$\{d\}_{-1} = \frac{\Delta t^2}{2}\{\ddot{d}\}_0, \qquad \{\ddot{d}\}_0 = [M]^{-1}\{F\}_0. \tag{3.17}$$

3.6. Critical Time Step

Central difference is stable only if every mode satisfies $\omega\,\Delta t < 2$ (Appendix C), so the highest natural frequency of the mesh sets the limit (Cook et al., 2001, ch. 11):

$$\Delta t_{cr} = \frac{2}{\omega_{max}}. \tag{3.18}$$

Estimates from element size and wave speed ignore the hoop stiffness and the mass matrix, so ω_{max} is found from the assembled model, following Groenewald (2024), by solving

$$[K]\{\phi\} = \omega^2 [M]\{\phi\}, \tag{3.19}$$

where $\{\phi\}$ is the mode shape. The problem is solved for the same constrained system that is stepped in time, so that Δt_{cr} is the limit of the system actually integrated. It is recalculated for every mesh and mass matrix, and $\Delta t = 0.9\,\Delta t_{cr}$ is used as a safety margin.

3.7. Mesh, Assembly and Implementation

The half-section is divided into a structured mesh of rectangular Q4 elements, with n_r elements across the radius and n_z along the length. Nodes are numbered row by row from the loaded face (Figure 3.4), and each element's nodes are listed anticlockwise to keep the Jacobian positive. The element matrices are assembled with the direct stiffness method (Cook et al., 2001) and stored as dense arrays, whose memory grows with the square of the number of degrees of freedom and limited the mesh size on the 16 GB machine used.

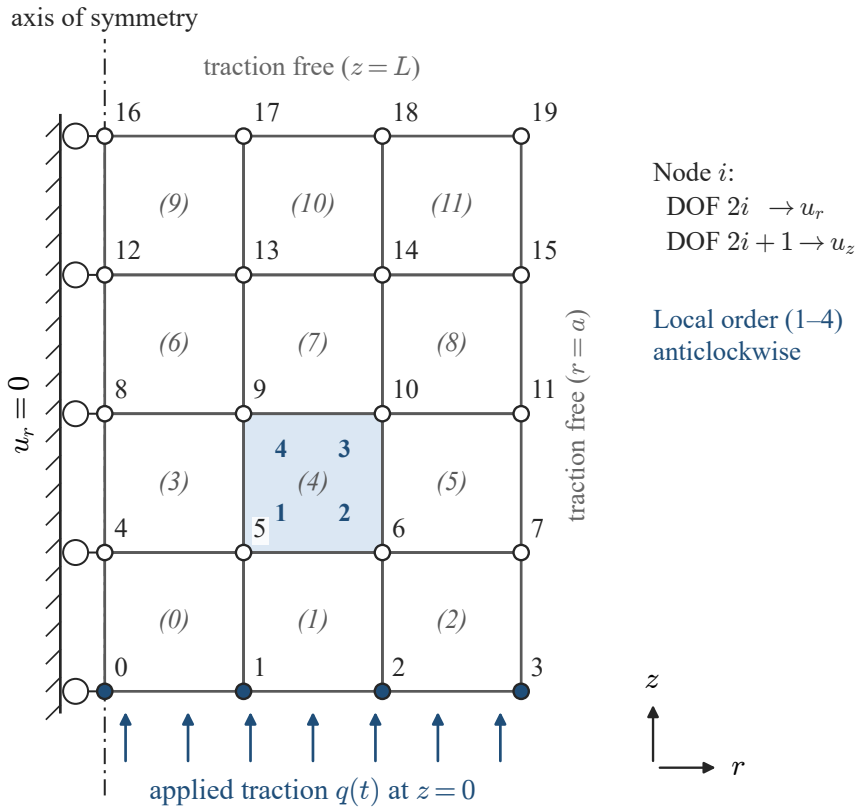


Figure 3.4: Example 3×4 mesh showing the global node numbers, element numbers (in brackets) and the anticlockwise local node order of one element. Nodes are numbered row by row from the loaded face, so the loaded nodes (filled) are always the first $n_r + 1$ nodes, where n_r is the number of radial elements.

The solver was written in Python with NumPy and SciPy. All finite element functions sit in one module called by a separate script for each test, so every test uses the same solver (Figure 3.5).

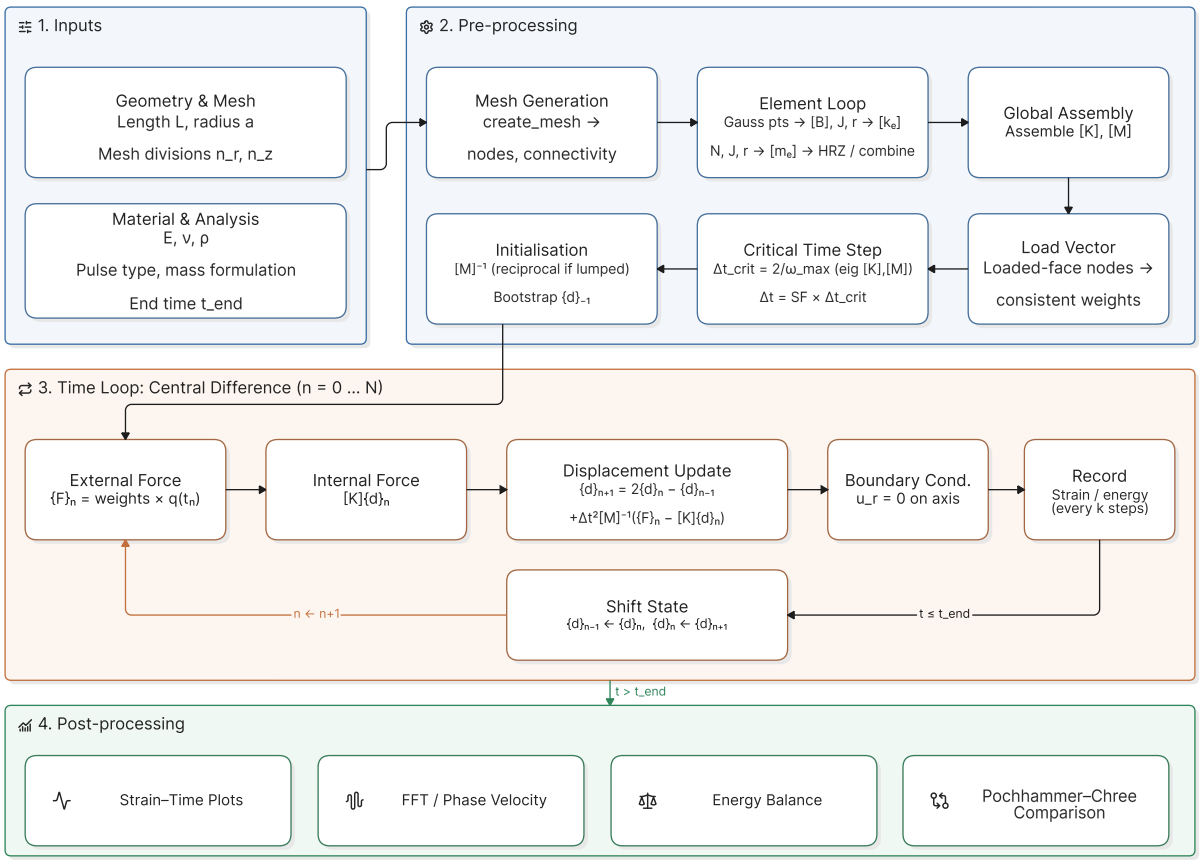


Figure 3.5: Program flow of the explicit solver. All matrices and the time step are calculated once before the time loop, which repeats the central difference update until the end time is reached. Diagram drawn with an AI-assisted tool (Eraser AI) from the author's specification.

3.8. Output

The main output is the axial strain against time at a gauge position z_g near the surface, comparable with strain gauge measurements. Except where a gauge length is used (Section 3.9.4), the strain is recovered from $\{\varepsilon\} = [B]\{d\}$ at the centre of the element nearest the gauge, where Q4 strains are most accurate (Cook et al., 2001), so the measurement sits at $r = a - \Delta r/2$. The kinetic and strain energies are

$$T = \tfrac{1}{2}\{\dot{d}\}^T[M]\{\dot{d}\}, \qquad U = \tfrac{1}{2}\{d\}^T[K]\{d\}. \tag{3.20}$$

3.9. Verification

Verification checks that the equations are solved correctly, while validation (Section 3.10) checks that they describe the real bar (Oberkampf and Roy, 2010). The tests build up from single matrices to the full solver: the patch test and nodal load check test $[K]$ and $\{F\}$, the energy, momentum and convergence tests check the time stepping, and the reflection and square-pulse tests check how waves behave on the mesh (Table 3.2).

Table 3.2: Test parameters, with the mesh given as radial × axial elements. GWS: 500 kHz, 4 cycles; Hann: 505 kHz, 5 cycles; square: 5 μs. The strain–time validation uses a Gaussian-windowed cosine with a 1 MHz carrier and a 1.5 μs half-width, and the phase-velocity test a 1 μs Hann pulse. All verification tests are run with lumped, consistent and combined mass, and the validation tests with lumped mass. All runs use $\Delta t = 0.9\,\Delta t_{cr}$, computed separately for each mesh and mass matrix.

Test	Radius [mm]	Length [mm]	Mesh	Pulse	Gauge [mm]
Patch test	2	2	4×4	–	–
Nodal loads	2	10	5×20	–	–
Energy and momentum	2	80	12×200	GWS, 100 MPa	–
Convergence	2	50	4×40–640	GWS, 100 MPa	20
Reflection	0.5	100	5×450	Hann, 200 MPa	70
Square pulse	2	80	4×200	Square, 100 MPa	50
Phase velocity	1	100	10×75–300	Hann, 200 MPa	20, 50
Strain–time	1	100	10×300	Gaussian, 200 MPa	60

3.9.1. Patch Test

The patch test checks that a group of elements reproduces a state of constant strain exactly, regardless of element shape, which is necessary for convergence (Cook et al., 2001). A constant shear state would need a body force to balance the τ_{rz}/r term in axial equilibrium, so the field

$$u_r = \varepsilon_{r0}\, r, \qquad u_z = \varepsilon_{z0}\, z + u_{z0} \tag{3.21}$$

is used instead, combining a rigid axial translation u_{z0} with constant strains $\varepsilon_r = \varepsilon_\theta = \varepsilon_{r0}$ and $\varepsilon_z = \varepsilon_{z0}$. The radial term is essential, as it is the only part that exercises the hoop row of $[B]$. The field is prescribed on the boundary of a distorted patch (Figure 3.6), and the free, unloaded interior nodes are found from

$$[K_{ii}]\{d_i\} = -[K_{ib}]\{d_b\}, \tag{3.22}$$

where the subscripts i and b denote the interior and boundary nodes. If the element is correct, the stress error at every Gauss point,

$$e = \frac{\|\{\sigma\} - [E]\{\varepsilon_0\}\|}{\|[E]\{\varepsilon_0\}\|}, \tag{3.23}$$

where $\{\varepsilon_0\}$ holds the prescribed strains, is at round-off level.

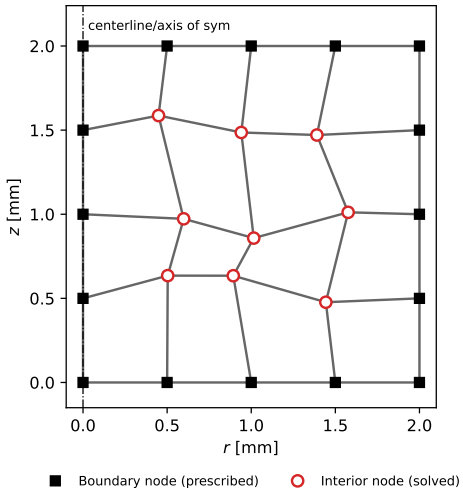


Figure 3.6: Patch test mesh: 2×2 mm, 4×4 elements, with interior nodes moved by up to 30 % of the element size. Boundary nodes are prescribed from Equation (3.21) ($\varepsilon_{r0} = 5 \times 10^{-4}$, $\varepsilon_{z0} = 10^{-3}$, $u_{z0} = 0.01$ mm).

3.9.2. Consistent Nodal Loads

An error in the load vector would distort the pulse from the start and be hard to separate from genuine dispersion. Setting $\varepsilon_{r0} = -\nu\,\varepsilon_{z0}$ in Equation (3.21) gives uniaxial stress $\sigma_z = E\varepsilon_{z0}$, and the nodal reactions are

$$\{R\} = [K]\{d\}. \tag{3.24}$$

Since the patch test confirms $[K]$, any mismatch can only come from the loads. The reactions on each end face should match Equation (3.13) with $q = E\varepsilon_{z0}$ node by node, summing to $\pi a^2 E\varepsilon_{z0}$, with zero reaction at every other node.

3.9.3. Energy and Momentum Balance

With no damping and no axial restraint, all the work done by the traction must remain in the bar, and its axial momentum must equal the applied impulse. The standard energy check for explicit codes is (Belytschko et al., 2000)

$$T + U = W, \qquad W = \int_0^t \{F\}^T \{\dot{d}\}\, dt. \tag{3.25}$$

Central difference naturally produces velocities halfway between time steps, so on-step velocities do not balance exactly. The energy is therefore calculated in the half-step form that the scheme conserves exactly (Hughes, 1987), and the balance should hold to round-off (Appendix D). Before the first reflection, T and U should also be approximately equal, as for any wave travelling in one direction. Since the internal forces cancel, only the traction can change the axial momentum, which should equal the applied impulse to round-off.

3.9.4. Mesh and Mass-Matrix Convergence

This test checks that the solution converges with mesh refinement and compares the rate for each mass matrix. Since the GWS pulse has a single dominant wavelength, resolution is described by the elements per wavelength, $\lambda/\Delta z$. The axial mesh is doubled from 40 to 640 elements with

four radial elements. To avoid sampling a different position on each mesh, the strain is averaged over a fixed 1 mm gauge length on the surface,

$$\varepsilon_g = \frac{u_z(a, z_g + g/2) - u_z(a, z_g - g/2)}{g}, \tag{3.26}$$

with u_z interpolated linearly between surface nodes. Each history is compared with the finest-mesh history of the same mass matrix using the relative L_2 error: the root-mean-square difference between the two, divided by the root-mean-square of the finest-mesh history. Since both the spatial and time discretisations are second-order accurate, the error should fall with a slope of about -2 against $\lambda/\Delta z$.

3.9.5. Single-Frequency Wave and Free-End Reflection

This test checks that a narrowband pulse travels at the expected speed and is reflected correctly. The stress at a free end must be zero, so the pulse returns inverted:

$$\varepsilon_{refl}(t) = -\varepsilon_{inc}\left(t - \frac{2(L - z_g)}{c_0}\right), \tag{3.27}$$

where $c_0 = \sqrt{E/\rho}$ (Graff, 1975, ch. 8). Since this relation assumes no dispersion, the Hann pulse is used in a thin bar ($a/\lambda \approx 0.05$), with the gauge far enough from the end that the incident and reflected pulses do not overlap. The time shift that best aligns them gives the wave-speed error, and the remaining difference shows how well shape and amplitude are preserved. Small errors are expected, as the sign flip is exact only in one dimension.

3.9.6. Square Pulse and Mass-Matrix Dispersion

The sharp edges of a square pulse contain a wide range of frequencies, showing how each mass matrix treats those the mesh cannot represent. With lumped mass every frequency travels at or below c_0, so the ringing trails each edge; with consistent mass most travel faster, so oscillations are also expected ahead of it (Cook et al., 2001, ch. 11). Poisson's ratio is set to zero, so the bar has no physical dispersion and any signal before z_g/c_0 is numerical. The solution is then uniform across the radius, so a coarse radial mesh lets the axial element size govern the time step. The gauge stress is compared with the ideal one-dimensional pulse.

3.10. Validation

The solver is validated against Pochhammer–Chree theory and the results of Govender et al. (2006) in three steps: the dispersion curves, the phase velocity measured from the model, and the strain–time response of a complete pulse.

3.10.1. Pochhammer–Chree Dispersion Curves

The Pochhammer–Chree frequency equation (Section 2.2) is used in the form given by Bancroft (1941) and adopted by Govender et al. (2006):

$$(x-1)^2\,\phi(ha) - (\beta x - 1)\,[x - \phi(ka)] = 0, \tag{3.28}$$

where

$$\beta = \frac{1-2\nu}{1-\nu}, \qquad x = (1+\nu)\left(\frac{c_p}{c_0}\right)^2, \qquad \gamma = \frac{2\pi}{\lambda},$$
$$h = \gamma\sqrt{\beta x - 1}, \qquad k = \gamma\sqrt{2x-1}, \qquad \phi(y) = \frac{y\,J_0(y)}{J_1(y)}, \tag{3.29}$$

and J_0 and J_1 are Bessel functions of the first kind, replaced by the modified functions I_0 and I_1 when h or k is imaginary. These symbols follow Bancroft (1941) and are local to this equation. Govender et al. (2006) print ϕ without the leading factor y; the form above is the one that reproduces their curves. For each a/λ, the roots x, one per mode, are found by scanning for sign changes and refining by bisection. The curves are compared with those of Govender et al. (2006) for the same bar; the first mode should start at c_0 and fall towards the Rayleigh wave speed.

3.10.2. Phase Velocity

The phase velocity of the model is measured directly and compared with the first mode. A short Hann pulse excites a wide range of frequencies, and the axial strain is recorded at two surface stations, z_1 and z_2, before the far-end reflection arrives. Since each frequency travels between the stations at its own phase velocity, the phase difference $\Delta\phi$ between the two spectra gives

$$c_p(f) = \frac{2\pi f\,(z_2 - z_1)}{\Delta\phi(f)}. \tag{3.30}$$

Only frequencies where the input spectrum exceeds 2 % of its peak, and below the second-mode cutoff, are used, so that the phase is reliable and a single mode is present. A 1 mm radius bar with the material of Govender et al. (2006) is used: since the solution depends only on a/λ, the results scale to their 10 mm bar, while the smaller bar needs fewer elements. Three axial meshes are run with lumped mass.

3.10.3. FEA against Pochhammer–Chree Prediction

The final check compares a complete dispersed pulse, following Section 3.3 of Govender et al. (2006). A Gaussian-windowed cosine with a 1 MHz carrier, equivalent to 100 kHz in their 10 mm bar, is applied at $z = 0$, and the strain is recorded at a surface gauge. The input strain, $\varepsilon(0,t) = -q(t)/E$ (negative in compression), is transformed into the frequency domain, and each component is delayed by its travel time at the first-mode phase velocity:

$$\hat{\varepsilon}(z_g,\omega) = \hat{\varepsilon}(0,\omega)\,e^{-i\omega z_g/c_p(\omega)}. \tag{3.31}$$

An inverse transform gives the predicted strain, with components above 1.8 MHz, where the spectrum is negligible, discarded. The first mode suffices because the pulse energy lies below the second-mode cutoff. For reference, the input is also shifted by z_g/c_0 to show its arrival without dispersion. The FEA should match the predicted arrival time and the trailing oscillations, with small differences since the prediction ignores higher modes and the radial variation of strain.

4

Results and Discussion

This chapter presents the results of the verification and validation tests in the order they were described, with the setup for each test given in Table 3.2. All verification tests were run with the lumped, consistent and combined mass matrices, while the validation uses lumped mass only. The chapter closes with a discussion of what the tests reveal about the mass matrices, the time step and the comparison with Govender et al. (2006).

4.1. Verification

4.1.1. Patch Test

Figure 4.1 shows the stress error in each element of the distorted patch from Figure 3.6. Every element reproduces the exact stress field, with errors no larger than about 10^{-14}, which corresponds to numerical round-off. Even the most distorted element, near the centre of the patch, shows no meaningful error. Since the prescribed field contains constant radial, axial and hoop strain, the patch test confirms that all three are captured exactly regardless of element shape. In particular, it confirms that the hoop strain term $\varepsilon_\theta = u_r/r$ in $[B]$ and the axisymmetric $[E]$ are implemented correctly, which a plane-stress or plane-strain patch test would not check.

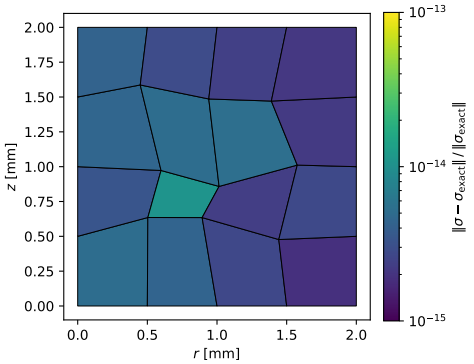


Figure 4.1: Relative stress error $\|\sigma - \sigma_{\text{exact}}\|/\|\sigma_{\text{exact}}\|$ per element for the distorted patch test. All errors are at round-off level ($< 10^{-13}$), so the patch test is passed.

4.1.2. Consistent Nodal Loads

Table 4.1 compares the reactions on the loaded face, $\{R\} = [K]\{d\}$ for a uniform axial strain, with the consistent nodal loads of Equation (3.13). Every node matches its analytical load, and the total equals the end force $\pi a^2 E\varepsilon_{z0}$ to all six decimal places. The loads at the interior nodes grow in proportion to radius, because nodes further from the axis support a larger ring of material, while the axis and outer nodes each receive the share of a single edge. An error in how this radial weighting is applied would immediately distort both the distribution and the total. The opposite end face shows the same loads acting in the reverse direction, and every other reaction in the bar is effectively zero. This confirms that the load is correctly distributed across the face as well as correct in total.

Table 4.1: Axial nodal reactions on the loaded face for the nodal load test (Table 3.2), with $E\varepsilon_{z0} = 200$ MPa. All values agree to within 10^{-12} %; the far face gives identical magnitudes.

r [mm]	FEA [N]	Analytical [N]
0.0	33.5103	33.5103
0.4	201.0619	201.0619
0.8	402.1239	402.1239
1.2	603.1858	603.1858
1.6	804.2477	804.2477
2.0	469.1445	469.1445
Total	2513.2741	2513.2741

4.1.3. Energy and Momentum Conservation

Figure 4.2 shows the energy history for the lumped mass matrix; the consistent and combined matrices give identical curves at this scale. The total energy $T + U$ follows the external work while the pulse is applied and stays constant once the load returns to zero, as expected for an undamped, traction-free bar. Before the pulse first reaches the far end, kinetic and strain energy are approximately equal, as expected for a wave travelling in one direction (Graff, 1975, ch. 1). From the first reflection onwards, the two exchange energy rapidly wherever the incident and reflected waves overlap. Because dispersion has spread the pulse into a long train by then, this exchange continues for most of the remaining run, while the total stays unchanged.

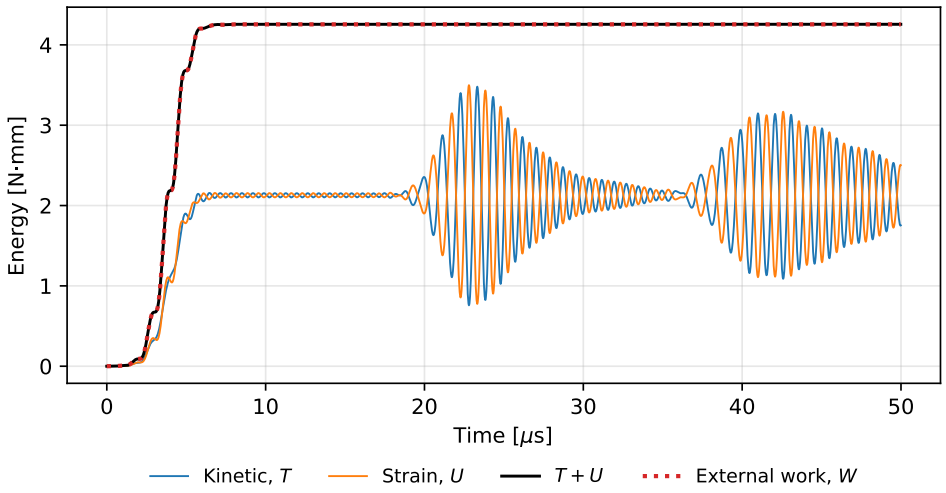


Figure 4.2: Energy components for lumped mass (energy test case, Table 3.2). $T + U$ equals the external work throughout, and $T \approx U$ before the first reflection.

Figure 4.3 shows how closely the total energy matches the work done when it is calculated in the half-step form of Appendix D. For all three mass matrices the difference is around 10^{-14} of the work done, which is the level of numerical round-off. It fluctuates around zero rather than growing in one direction, and the reflections leave no mark on it, so the solver neither creates nor loses energy at any point in the run.

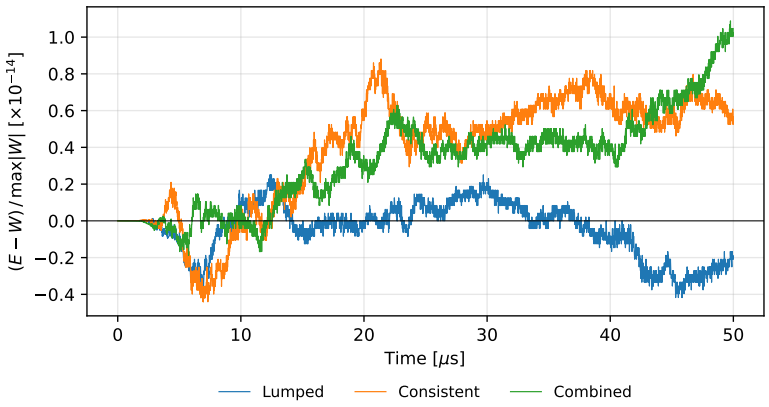


Figure 4.3: Energy balance error, $(T + U - W)/\max|W|$, with T and U calculated in the half-step form of Appendix D. For all three mass matrices the error stays within $\pm 1 \times 10^{-14}$.

Momentum behaves in the same way. The axial momentum of the bar matches the total impulse applied at the loaded face to within 1×10^{-13} of the peak impulse, again for all three mass matrices. Together, the energy and momentum results show that the central difference integration is implemented correctly.

4.1.4. Mesh and Mass-Matrix Convergence

Figure 4.4a shows the gauge strain for each level of refinement, using the combined mass matrix. The curves are almost indistinguishable, so even the coarsest mesh, with about eight elements per wavelength, captures the arrival time, shape and amplitude of the pulse. The small differences

that remain appear in the final oscillations, which are the most affected by dispersion.

Figure 4.4b shows that all three mass matrices converge, at close to second order or faster on the finest meshes. Lumped mass is the most accurate from 16 elements per wavelength onwards, reaching 0.45 % at 32 elements per wavelength, against 1.9 % for combined mass and 3.9 % for consistent mass, nearly nine times the lumped error. Only on the coarsest mesh is the combined matrix more accurate. The lumped error also falls faster than second order on the coarser meshes, with slopes of -2.8 and -3.0 before easing to -2.5. On the coarse meshes the radial elements set the time step, so the axial Courant number $c_0\Delta t/\Delta z$ is only 0.3–0.6; once the axial elements become the smaller dimension it settles at about 0.77. The lumped spatial error and the time-integration error cancel more completely as this number approaches one (Section 4.3.2), so lumped mass gains accuracy from the rising Courant number as well as from the smaller elements.

The peak gauge strain shows the same behaviour. For all three mass matrices it settles to within about 0.5 % of the finest-mesh value by 80 axial elements. On the coarsest mesh the consistent peak is 8.2 % low, against about 5 % for lumped and combined mass. Coarsening the radial mesh from four elements to two changes the lumped peak by at most 0.2 %, but raises the consistent and combined peaks by 1.1–1.6 %. For lumped mass, accuracy is therefore controlled almost entirely by the axial element size, while the consistent and combined matrices are also sensitive to the radial mesh.

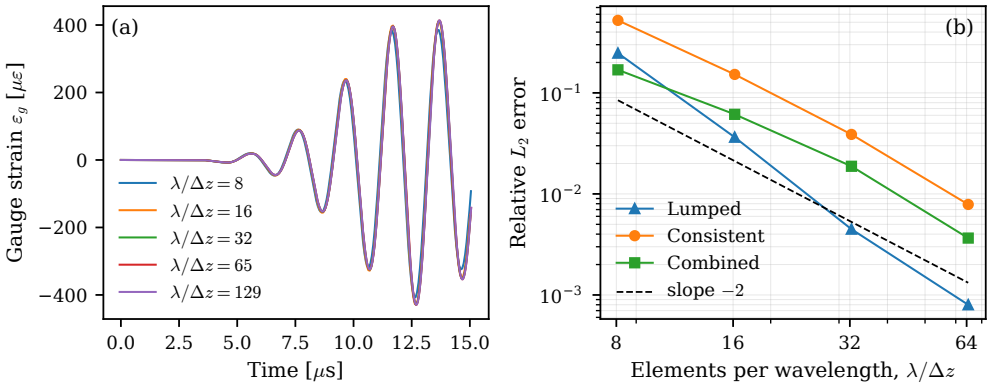


Figure 4.4: Mesh convergence with four radial elements (Table 3.2). (a) Gauge strain for each axial refinement, combined mass. (b) Relative L_2 error against the finest mesh ($\lambda/\Delta z \approx 129$) of the same mass matrix.

4.1.5. Single-Frequency Wave and Free-End Reflection

Figure 4.5 overlays the incident pulse with the inverted and shifted reflection. The free end reflects the pulse with the correct sign, shape and amplitude, and the reflection arrives only slightly later than predicted. Table 4.2 shows that, for all three mass matrices, this delay corresponds to a wave speed less than 0.7 % below c_0. This is close to what the remaining physical dispersion predicts. A narrowband pulse travels at the group velocity, which for a thin bar is approximately $c_g/c_0 \approx 1 - 3\nu^2\pi^2(a/\lambda)^2$ (Graff, 1975, ch. 2), or about 0.7 % below c_0 at $a/\lambda = 0.05$. The delay is therefore mainly a property of the bar rather than an error in the model. The smaller differences between the mass matrices, of up to 0.2 %, are their numerical

dispersion: consistent mass gives the fastest wave, as expected. Once the delay is removed, the two pulses differ by at most 2.2 % of the peak strain. The free end therefore reflects the pulse correctly, and the test is passed.

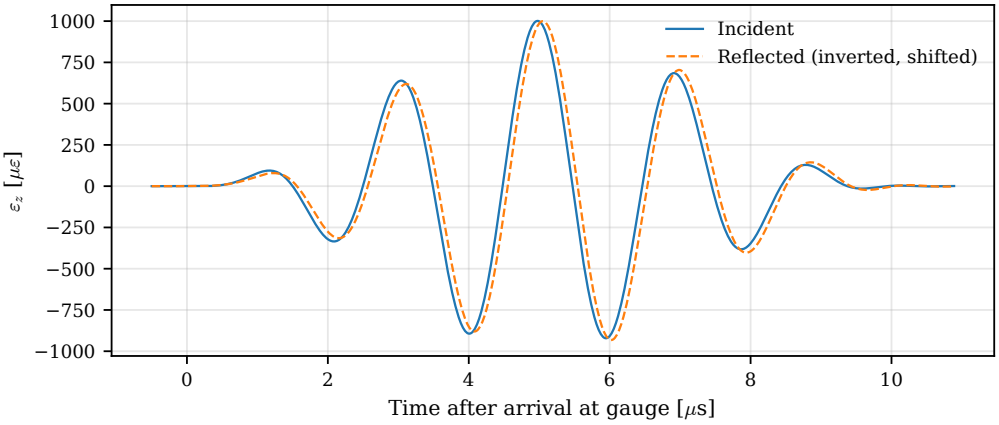


Figure 4.5: Incident and reflected strain at the gauge for lumped mass (Table 3.2). The reflected pulse is inverted and shifted by $2(L - z_g)/c_0$; the two overlap almost exactly.

Table 4.2: Reflection results for each mass matrix: wave-speed error over the extra travel distance, and the largest remaining difference once the pulses are aligned.

Mass matrix	Speed below c_0 [%]	Difference after alignment [%]
Lumped	0.64	2.17
Consistent	0.44	1.02
Combined	0.63	1.50

4.1.6. Square Pulse and Mass-Matrix Dispersion

Figure 4.6 shows the behaviour expected in Section 3.9.6. With lumped mass, the ringing trails behind each edge of the pulse, with an overshoot of about 25 % of the applied stress, and nothing arrives significantly early. With consistent mass, oscillations appear about 4.3 μs ahead of the front as well as behind it, and the precursor of the falling edge makes the oscillations grow towards the end of the plateau. In both cases the main front arrives at the ideal time z_g/c_0. Each mass matrix therefore produces its known dispersion signature, so the test is passed and both matrices are confirmed to be implemented correctly.

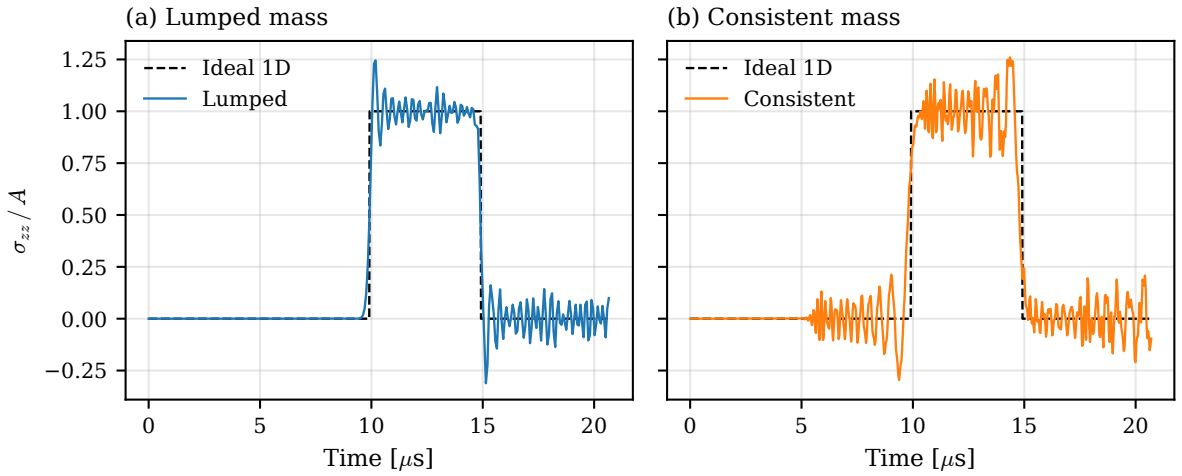


Figure 4.6: Axial stress at $z_g = 50$ mm for the square-pulse test (Table 3.2), normalised by the applied amplitude. (a) Lumped mass: ringing trails the front. (b) Consistent mass: oscillations arrive ahead of the front.

4.2. Validation

With the solver verified, its predictions are now compared with Pochhammer–Chree theory and the results of Govender et al. (2006), to assess whether the model captures the physical dispersion in the bar.

4.2.1. Pochhammer–Chree Dispersion Curves

The dispersion curves from the implementation reproduce those of Govender et al. (2006) (Figure 4.7). The first mode starts at c_0 for long wavelengths and falls smoothly towards the Rayleigh wave speed as the frequency rises, while each higher mode appears at its cutoff frequency with a very high phase velocity and falls towards the shear wave speed. The shape of each curve and the frequencies at which the higher modes appear agree with the published figure, confirming that the frequency equation is implemented correctly and can be used as the reference for the finite element results that follow.

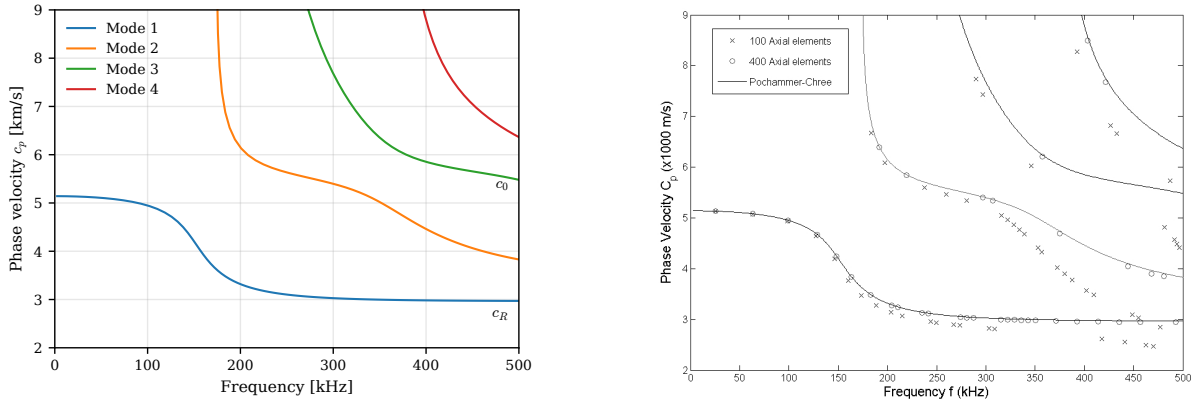


Figure 4.7: Pochhammer–Chree dispersion curves for a 20 mm diameter steel bar ($\nu = 0.26$): calculated in this project (left) and reproduced from Govender et al. (2006, Figure 1) (right), where the solid lines are the Pochhammer–Chree solution and the markers are their finite element results.

4.2.2. Phase Velocity

Figure 4.8a shows that the measured phase velocity follows the first Pochhammer–Chree mode on all three meshes, with the coarser meshes falling below the curve at lower frequencies. Figure 4.8b shows why: the errors from all three meshes collapse onto a single curve when plotted against elements per wavelength, $\lambda/\Delta z$, so the error is controlled by mesh resolution. The error stays below 1 % down to about 15 elements per wavelength and grows rapidly below that. On the two finer meshes the measurement ends at about 145–150 kHz, because the input pulse contains little energy above this frequency; the coarsest mesh ends earlier, as it carries even less of the high-frequency content. Within the measured range, the model therefore reproduces the physical dispersion of the bar, with an accuracy set by the axial mesh resolution.

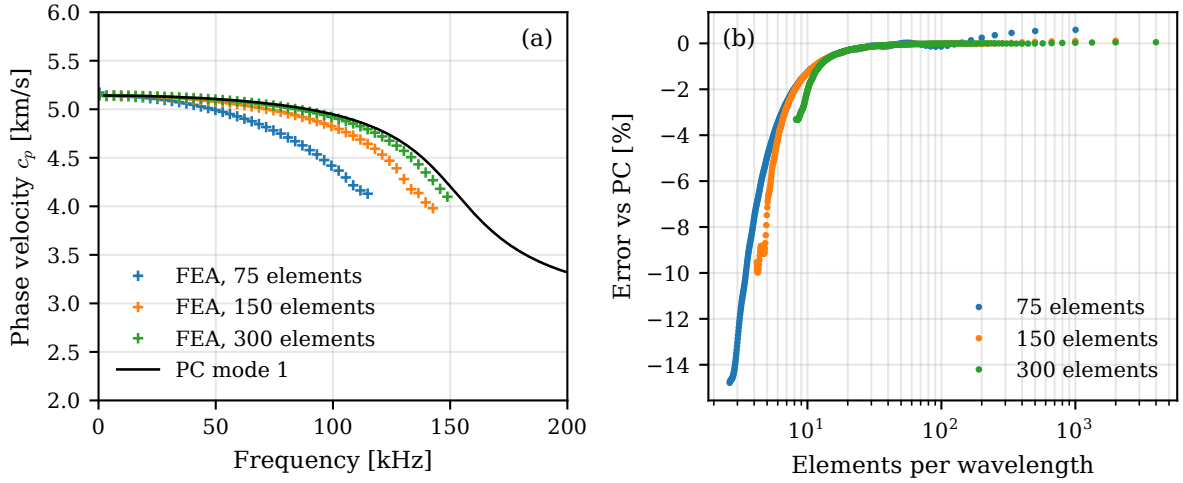


Figure 4.8: Phase velocity from the two-station method for three axial meshes (Table 3.2), with frequencies scaled to the 10 mm radius bar of Govender et al. (2006). (a) Measured phase velocity against the first Pochhammer–Chree mode. (b) Error against elements per wavelength: the three meshes collapse onto one curve.

4.2.3. FEA against Pochhammer–Chree Prediction

Figure 4.9 compares the strain at the gauge with the Pochhammer–Chree prediction and with the undispersed pulse shifted by z_g/c_0. The FEA pulse arrives later than the undispersed pulse and has spread out, with high-frequency oscillations trailing behind the main signal, and the prediction captures both effects. The FEA lags it by only 0.08 μs, which matches the phase-velocity error of about 0.7 % expected at this mesh resolution of around 15 elements per wavelength (Section 4.2.2). The FEA peak is about 8 % lower than predicted, and the normalised root-mean-square difference between the two signals is 12 % of the peak. These differences are consistent with the simplifications in the prediction, which uses only the first mode and assumes the strain is uniform across the bar. The model therefore reproduces the arrival time and dispersed shape of a complete pulse, and the validation is successful.

4.3. Discussion

4.3.1. Comparison of the Mass Matrices

All three mass matrices passed every verification test, but together the tests show clear differences in how accurately they model wave propagation. In the convergence study, lumped mass was

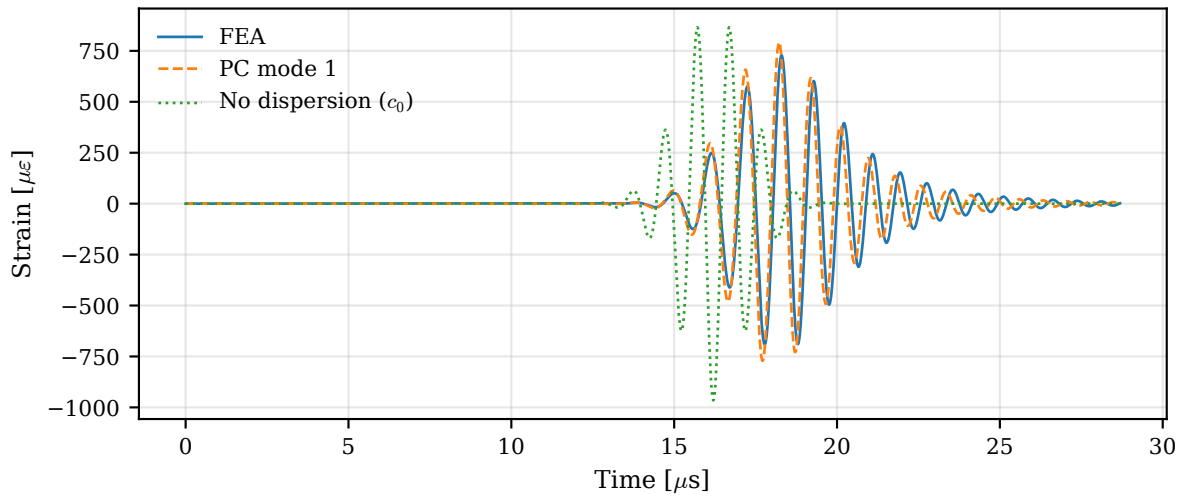


Figure 4.9: Axial strain at the gauge for the strain–time test (Table 3.2), compared with the first-mode Pochhammer–Chree prediction and the undispersed pulse arriving at z_g/c_0.

the most accurate on all but the coarsest mesh, with about a quarter of the combined error and a ninth of the consistent error at 32 elements per wavelength. In the square-pulse test, lumped mass kept all of its numerical error behind the front, while consistent mass produced a spurious signal ahead of it. Spatial discretisation with lumped mass makes high-frequency waves travel too slowly, while central difference time integration makes them travel too fast, so the two errors partly cancel. With consistent mass, both errors make waves travel too fast, so they add (Cook et al., 2001, ch. 11; Appendix C). The combined matrix removes most of the spatial error but leaves the time-integration error in place, so it gains less from the cancellation: it is the most accurate on the coarsest mesh, where the cancellation is weakest, but falls behind lumped mass as the mesh is refined. The consistent and combined matrices were also more sensitive to the radial mesh, likely because their off-diagonal terms couple the inertia of neighbouring nodes, so the distribution of mass across the radius depends on how finely it is meshed.

The square pulse also shows that consistent mass does not simply make every wave too fast. Ringing appeared behind each edge as well as ahead of it, because the energy of a wave travels at its group velocity rather than its phase velocity. With consistent mass, most frequencies have a group velocity above c_0, which produces the precursor, but the highest frequencies the mesh can carry have a group velocity close to zero and fall far behind the front. The usual statement that consistent mass leads and lumped mass trails therefore describes only the most visible part of the behaviour.

Lumped mass is also the cheapest option. Its inverse is diagonal, so each step needs only a division per degree of freedom rather than a dense matrix–vector product, and its critical time step is the largest of the three (Appendix C). Being the most accurate of the three on all but the coarsest mesh, insensitive to the radial mesh and the cheapest to run, lumped mass is the natural choice for explicit wave propagation, which agrees with common practice (Belytschko et al., 2000, ch. 6).

4.3.2. Time Step in Two Dimensions

The cancellation between the lumped mass and time-integration errors is complete only when a wave crosses exactly one element per time step. In one dimension, this condition coincides with the stability limit, so a well-chosen time step removes the dispersion almost entirely. In two dimensions it cannot be reached, because the critical time step depends on both element dimensions. In the square-pulse test, increasing the time step from 0.3 to $0.9\,\Delta t_{cr}$ reduced the lumped overshoot only modestly, from 31 % to 25 %, as the axial wave still crossed well under one element per step. The convergence runs show this dependence directly. With 40 axial elements, halving the number of radial elements almost doubled the lumped time step, from 7.6×10^{-8} to 1.5×10^{-7} s, because the radial element size was setting it. From 160 axial elements onwards, the same change made almost no difference, as the axial elements had become the smaller dimension. The rise in the axial Courant number over these first refinements is also why the lumped error fell faster than second order (Section 4.1.4). It is therefore refining the axial mesh, rather than adjusting the time step, that reduces dispersion.

During this work, the radial displacements on the axis were also found to affect the time step. If they are left unconstrained, the large hoop stiffness near $r = 0$ produces a spurious high-frequency mode that sets Δt_{cr}, even though the symmetry of the problem keeps these displacements at zero. Removing them from the system, in both the eigenvalue problem and the time stepping, ensures that the time step is set by the physical mesh instead. Without the constraint, the spurious mode rather than the mesh set the time step, so the measured convergence rates reflected that mode rather than the discretisation, which shows how easily an unphysical mode can distort a convergence study.

4.3.3. Comparison with Govender et al.

The validation results agree with Govender et al. (2006) on several independent points. Both studies reproduce the Pochhammer–Chree dispersion curves, and both find that, with lumped mass, accuracy is controlled by the axial rather than the radial element size. Their eigen-analysis showed that the first-mode error exceeded 1 % once the wavelength fell below about 17 axial element lengths, while the two-station measurements here give the same 1 % threshold at about 15 elements per wavelength. That two different solvers and two different ways of extracting the phase velocity reach almost the same requirement gives confidence in both results.

Govender et al. (2006) also found that explicit finite element results agreed well with the first-mode prediction for pulses below 100 kHz, but differed strongly at 200 kHz, where the second mode is excited. The strain–time comparison here used a carrier equivalent to 100 kHz in their bar and agreed well in arrival time and shape, consistent with their findings. Their reported limit of 150 kHz could not be tested directly, as the phase-velocity measurement here was limited by the bandwidth of the input pulse (Section 4.2.2). Extending the comparison to higher frequencies would require a shorter pulse and a finer mesh than the dense matrix storage currently allows.

5

Conclusions and Recommendations

5.1. Conclusions

The aim of this project was to develop an explicit finite element solver for slender bars using the central difference scheme, and to verify it through targeted tests, as part of a series working towards an open code for studying higher-mode dispersion in HPBs. This aim was met in full, and the project was extended to validate the solver against Pochhammer–Chree theory and the results of Govender et al. (2006).

The existing code was found unsuitable as a basis for explicit analysis: a coding error doubled its stiffness matrix, which had been offset by doubling the mass, its inputs were fixed arrays that had to be edited by hand to change the mesh, and it had no time integration. A new, modular solver was therefore written in Python for axisymmetric Q4 elements. Its formulation was derived in full, including the stiffness matrix, the lumped, consistent and combined mass matrices, the consistent nodal loads and the stability limit of the central difference scheme (Appendices A–E), with the critical time step found from the constrained eigenvalue problem.

The verification was complete. The patch test and nodal load check confirmed the stiffness matrix and load vector to round-off, energy and momentum were conserved to round-off for all three mass matrices, and the solution converged at second order or faster. The reflection test reproduced the free-end behaviour, with a small delay matching the group velocity of a thin bar, and the square-pulse test confirmed the dispersion signature of each mass matrix. Lumped mass was the most accurate on all but the coarsest mesh, with about a ninth of the consistent error at 32 elements per wavelength, and was insensitive to the radial mesh. Since its diagonal form also makes each step the cheapest and allows the largest stable time step, it is the most suitable choice for this application.

As an extension beyond the brief, the solver was validated in the first-mode regime. The Pochhammer–Chree dispersion curves of Govender et al. (2006) were reproduced, the measured phase velocity agreed with the first mode to within 1 % wherever there were at least about 15 elements per wavelength, close to their threshold of 17, and a complete dispersed pulse matched the first-mode prediction in arrival time and shape. The two-mode regime at 200 kHz was

not tested; extending the validation to the higher modes that motivate the series is the main recommendation of this work.

5.2. Limitations

The main limitation on extending this work to the higher modes is the efficiency of the code. All global matrices are stored as dense arrays, so memory grows with the square of the number of degrees of freedom. Each matrix then occupies about 2.5 GB at roughly 17 000 degrees of freedom, the practical limit on a 16 GB machine. Higher modes appear where wavelengths are only around 1.5 mm in the scaled bar, and need both at least 15–17 elements per wavelength and a bar long enough for the modes to separate in time, which corresponds to tens to hundreds of thousands of degrees of freedom. The current code therefore could not resolve the higher modes even with an extraction method able to separate them, and the two-station method used here measures only a single mode in any case. The remaining limitations are listed in Table 5.1.

5.3. Recommendations

The first priority for the series is to rewrite the matrix storage and solvers for efficiency, storing $[K]$ in sparse format, the lumped mass as a vector, and finding ω_{max} with a sparse eigenvalue solver. This would reduce the memory needed for multi-mode meshes from hundreds of gigabytes to under a gigabyte, and is a prerequisite for extending the validation into the multi-mode regime using the approach of Barr et al. (2020), with the 200 kHz two-mode result of Govender et al. (2006) as a natural target.

A further question is whether the time step can be increased beyond the critical value while keeping the solution stable, for example through mass scaling or subcycling, which adjust the highest frequencies or use different time steps in different parts of the mesh (Belytschko et al., 2000, ch. 6). Since both alter the high-frequency response that governs dispersion, their effect on accuracy would need to be checked against the results presented here. The responses to each limitation are given in Table 5.1, and further directions for the series in Table 5.2.

Table 5.1: Limitations of the present work and the recommended response to each.

Limitation	Recommendation
Dense matrix storage limits the mesh to roughly 17 000 degrees of freedom on a 16 GB machine.	Store $[K]$ in sparse format and lumped mass as a vector, and use a sparse eigenvalue solver for Δt_{cr}.
The two-station method can only measure a single mode, so validation is limited to the first mode.	Separate the modes by group velocity using windowed pulses on a long bar, as demonstrated by Barr et al. (2020), and reproduce the 200 kHz result of Govender et al. (2006).
The first-mode phase velocity was measured only up to about 150 kHz, set by the bandwidth of the input pulse, and the mesh gave only about 6 elements per wavelength near the second-mode cutoff.	Use a shorter input pulse together with a finer axial mesh, made possible by sparse storage, to extend the first-mode measurement accurately up to the second-mode cutoff.
The strain–time prediction assumes the strain is uniform across the section, taking the input as $-q(t)/E$ and ignoring its radial variation.	Add the amplitude correction of Tyas and Watson (2001) to the prediction.

Table 5.2: Recommended further work for the series.

Recommendation	Purpose
Find the phase velocity of the higher modes from an eigen-analysis of a bar with zero axial displacement at its ends, following Govender et al. (2006).	Check the higher-mode dispersion against Pochhammer–Chree theory without time integration, before attempting multi-mode wave propagation.
Compare the strain at several radii with the radial distribution predicted by Davies (1948).	Validate the cross-sectional variation of strain, which the present validation does not test.
Implement and compare alternative time integration schemes, such as Adams–Bashforth or Newmark.	Assess their accuracy and cost against central difference using the same verification tests.
Study stiffness-proportional damping.	Assess whether controlled damping can suppress high-frequency numerical noise without distorting the physical dispersion.
Investigate the radial-mesh sensitivity of the consistent and combined matrices, and the effect of the weighting β.	Explain the observed sensitivity, and establish whether a different weighting improves the combined matrix.
Study the effect of element aspect ratio.	Quantify how much accuracy improves when the axial element size governs Δt_{cr}.
Compare with experimental HPB measurements.	Confirm the model against physical data as well as theory.

6

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A

Axisymmetric Element Stiffness Matrix

This appendix derives the stiffness matrix $[k]$ of the four-node axisymmetric element. The isoparametric mapping, the Jacobian and the $[H][\Gamma][\partial N]$ chain follow the plane bilinear quadrilateral of Cloete (2020). Two changes make the element axisymmetric: a fourth strain row for the hoop strain, and the factor $2\pi r$ in the volume integral. The implementation is the function `compute_B_and_jacobian` in `fea_core.py`.

A.1. Strain energy

The stiffness matrix comes from the strain energy of the element (Cook et al., 2001, ch. 4). If the strains are written in terms of the nodal displacements as $\{\varepsilon\} = [B]\{d\}$, then

$$U_e = \frac{1}{2}\int_{V_e} \{\varepsilon\}^T [E]\{\varepsilon\}\, dV = \frac{1}{2}\{d\}^T [k]\{d\}, \qquad [k] = \int_{V_e} [B]^T [E][B]\, dV, \tag{A.1}$$

where, for a body of revolution, $dV = 2\pi r\, dr\, dz$. The rest of this appendix finds $[B]$ and $[E]$ and evaluates the integral.

A.2. Isoparametric mapping

Nodes are numbered anticlockwise in the (r, z) plane: node 1 at the lower-inner corner, node 2 lower-outer, node 3 upper-outer and node 4 upper-inner. The natural coordinate ξ runs in the radial direction and η in the axial direction, both from -1 to 1. The bilinear shape functions are

$$\begin{aligned} N_1 &= \tfrac{1}{4}(1-\xi)(1-\eta), \quad N_2 = \tfrac{1}{4}(1+\xi)(1-\eta), \\ N_3 &= \tfrac{1}{4}(1+\xi)(1+\eta), \quad N_4 = \tfrac{1}{4}(1-\xi)(1+\eta). \end{aligned} \tag{A.2}$$

The element is isoparametric: the same functions interpolate both the position and the displacement of any point,

$$r = \sum_{i=1}^{4} N_i r_i, \qquad z = \sum_{i=1}^{4} N_i z_i, \qquad \{u\} = \begin{Bmatrix} u_r \\ u_z \end{Bmatrix} = [N]\{d\}, \tag{A.3}$$

with

$$[N] = \begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \\ 0 & N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 \end{bmatrix}, \tag{A.4}$$

$$\{d\} = \{u_{r1}\ u_{z1}\ u_{r2}\ u_{z2}\ u_{r3}\ u_{z3}\ u_{r4}\ u_{z4}\}^T. \tag{A.5}$$

A.3. Strain–displacement matrix

The four strain components, in the order used throughout the report, are

$$\{\varepsilon\} = \begin{Bmatrix} \varepsilon_r \\ \varepsilon_z \\ \gamma_{rz} \\ \varepsilon_\theta \end{Bmatrix} = \begin{Bmatrix} \partial u_r/\partial r \\ \partial u_z/\partial z \\ \partial u_r/\partial z + \partial u_z/\partial r \\ u_r/r \end{Bmatrix}. \tag{A.6}$$

The first three are the same as in a plane problem. The hoop strain is new. A ring of material at radius r that moves outward by u_r has its circumference changed from $2\pi r$ to $2\pi(r + u_r)$, so

$$\varepsilon_\theta = \frac{2\pi(r + u_r) - 2\pi r}{2\pi r} = \frac{u_r}{r}. \tag{A.7}$$

Because ε_θ contains no derivative, it is built separately from the other three rows.

Derivative rows. The shape functions are written in (ξ, η), but the strains need derivatives with respect to (r, z). The derivatives are therefore taken in natural coordinates first and then converted with the chain rule. Differentiating Equation (A.3) gives

$$\{u_{,\xi}\} = \begin{Bmatrix} \partial u_r/\partial \xi \\ \partial u_r/\partial \eta \\ \partial u_z/\partial \xi \\ \partial u_z/\partial \eta \end{Bmatrix} = [\partial N]\{d\}, \tag{A.8}$$

$$[\partial N] = \frac{1}{4}\begin{bmatrix} -(1-\eta) & 0 & (1-\eta) & 0 & (1+\eta) & 0 & -(1+\eta) & 0 \\ -(1-\xi) & 0 & -(1+\xi) & 0 & (1+\xi) & 0 & (1-\xi) & 0 \\ 0 & -(1-\eta) & 0 & (1-\eta) & 0 & (1+\eta) & 0 & -(1+\eta) \\ 0 & -(1-\xi) & 0 & -(1+\xi) & 0 & (1+\xi) & 0 & (1-\xi) \end{bmatrix}. \tag{A.9}$$

By the chain rule, the natural derivatives of u_r are related to its physical derivatives through the Jacobian matrix $[J]$ (Cloete, 2020):

$$\begin{Bmatrix} \partial u_r/\partial \xi \\ \partial u_r/\partial \eta \end{Bmatrix} = [J] \begin{Bmatrix} \partial u_r/\partial r \\ \partial u_r/\partial z \end{Bmatrix}, \tag{A.10}$$

where

$$[J] = \begin{bmatrix} \partial r/\partial \xi & \partial z/\partial \xi \\ \partial r/\partial \eta & \partial z/\partial \eta \end{bmatrix} = \begin{bmatrix} \partial N_1/\partial \xi & \cdots & \partial N_4/\partial \xi \\ \partial N_1/\partial \eta & \cdots & \partial N_4/\partial \eta \end{bmatrix} \begin{bmatrix} r_1 & z_1 \\ r_2 & z_2 \\ r_3 & z_3 \\ r_4 & z_4 \end{bmatrix}. \tag{A.11}$$

The same relation holds for u_z. Inverting $[J]$ for both displacement components at once gives

$$\{u_{,r}\} = \begin{Bmatrix} \partial u_r/\partial r \\ \partial u_r/\partial z \\ \partial u_z/\partial r \\ \partial u_z/\partial z \end{Bmatrix} = [\Gamma]\{u_{,\xi}\}, \qquad [\Gamma] = \begin{bmatrix} [J]^{-1} & [0] \\ [0] & [J]^{-1} \end{bmatrix}. \tag{A.12}$$

The determinant $|J|$ is the area scale factor between the two coordinate systems, $dr\,dz = |J|\,d\xi\,d\eta$ (Cloete, 2020). Finally, $[H]$ picks out and combines the physical derivatives into ε_r, ε_z and γ_{rz}:

$$\begin{Bmatrix} \varepsilon_r \\ \varepsilon_z \\ \gamma_{rz} \end{Bmatrix} = [H]\{u_{,r}\}, \qquad [H] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}. \tag{A.13}$$

Hoop row. Substituting $u_r = \sum N_i u_{ri}$ into $\varepsilon_\theta = u_r/r$ gives the fourth row directly,

$$[B_\theta] = \frac{1}{r}\begin{bmatrix} N_1 & 0 & N_2 & 0 & N_3 & 0 & N_4 & 0 \end{bmatrix}, \tag{A.14}$$

where $r = \sum N_i r_i$ is the radius of the point at which $[B]$ is evaluated. In the solver this is always a Gauss point (or the element centre for strain output), never a node, so $r > 0$ even for elements that touch the axis.

Assembled $[B]$. Stacking the two parts gives the 4×8 strain–displacement matrix

$$\{\varepsilon\} = [B]\{d\}, \qquad [B] = \begin{bmatrix} [H][\Gamma][\partial N] \\ [B_\theta] \end{bmatrix}. \tag{A.15}$$

A.4. Elasticity matrix

In an axisymmetric body the radial, axial and hoop directions are all principal directions of normal strain, and the shear strains $\gamma_{r\theta}$ and $\gamma_{z\theta}$ are zero by symmetry. The hoop strain therefore plays the role of the third normal strain in the 3D form of Hooke's law (Cloete, 2020), and τ_{rz} is the only shear stress. With the strain order of Equation (A.6),

$$\begin{Bmatrix} \sigma_r \\ \sigma_z \\ \tau_{rz} \\ \sigma_\theta \end{Bmatrix} = [E]\{\varepsilon\}, \qquad [E] = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1 & \frac{\nu}{1-\nu} & 0 & \frac{\nu}{1-\nu} \\ \frac{\nu}{1-\nu} & 1 & 0 & \frac{\nu}{1-\nu} \\ 0 & 0 & \frac{1-2\nu}{2(1-\nu)} & 0 \\ \frac{\nu}{1-\nu} & \frac{\nu}{1-\nu} & 0 & 1 \end{bmatrix}. \tag{A.16}$$

The shear term reduces to the shear modulus, $E/[2(1+\nu)]$.

A.5. Numerical integration

Mapping Equation (A.1) to natural coordinates and applying Gauss quadrature in both directions gives

$$[k] = \int_{-1}^{1}\int_{-1}^{1} [B]^T[E][B]\, 2\pi r\, |J|\, d\xi\, d\eta \approx \sum_{i=1}^{3}\sum_{j=1}^{3} w_i w_j \Big([B]^T[E][B]\, 2\pi r\, |J|\Big)_{\xi_i,\,\eta_j}, \tag{A.17}$$

where $[B]$, r and $|J|$ are all evaluated at each Gauss point. The three-point rule uses $\xi_i = -\sqrt{3/5}$, 0, $\sqrt{3/5}$ with weights $w_i = 5/9$, $8/9$, $5/9$, and integrates polynomials up to degree five exactly (Gerald and Wheatley, 1989, §4.12).

Why three points. For a rectangular element, $[J]$ is constant and most terms of the integrand are polynomials in ξ and η. The exception is the product of two hoop rows, which contributes terms of the form $N_i N_j |J|/r$. Because of the $1/r$, this term is not a polynomial and no Gauss rule integrates it exactly. The error is largest in the elements next to the axis, where r varies most across the element relative to its mean value. The 3×3 rule was used to keep this error small. The consistent mass integrand, $\rho N_i N_j\, 2\pi r\, |J|$, is at most of degree four in each direction, so the same 3×3 rule integrates $[M]$ exactly.

Since the problem is linear and the mesh does not move, $[k]$ is computed once per element and assembled into $[K]$ before time stepping begins.

B

Consistent Nodal Loads

This appendix derives the nodal forces that are equivalent to the uniform axial traction $q(t)$ on the loaded face $z = 0$. The approach follows Cloete (2020): the loads are chosen so that they do the same work as the distributed traction.

B.1. Work-equivalent loads

The external work term in the total potential energy of an element with surface traction $\{\Phi\}$ on its boundary S is

$$W_S = \int_S \{u\}^T \{\Phi\}\, dS. \tag{B.1}$$

Along an element edge on $z = 0$, between radii r_1 and r_2 with $h = r_2 - r_1$, the axial displacement varies linearly:

$$u_z(r) = [N]\{u_z\}, \qquad [N] = \begin{bmatrix} \dfrac{r_2 - r}{h} & \dfrac{r - r_1}{h} \end{bmatrix}. \tag{B.2}$$

If the traction is interpolated with the same functions, $q(r) = [N]\{q\}$ with nodal values $\{q\} = \{q_1 \;\; q_2\}^T$, and the face is a ring so that $dS = 2\pi r\, dr$, then

$$W_S = \{u_z\}^T \{Q\}, \qquad \{Q\} = \left(\int_{r_1}^{r_2} [N]^T [N]\, 2\pi r\, dr \right) \{q\}. \tag{B.3}$$

$\{Q\}$ is the vector of consistent (work-equivalent) nodal loads.

B.2. Evaluating the integral

With the substitution $s = r - r_1$, so that $r = r_1 + s$, $r_2 - r = h - s$ and s runs from 0 to h, each entry is a polynomial integral. For example,

$$\int_{r_1}^{r_2} N_1^2\, 2\pi r\, dr = \frac{2\pi}{h^2} \int_0^h (r_1 + s)(h - s)^2\, ds = \frac{2\pi}{h^2} \left(\frac{r_1 h^3}{3} + \frac{h^4}{12} \right) = \frac{\pi h}{6} (4r_1 + h). \tag{B.4}$$

The other entries follow in the same way, giving

$$\{Q\} = \frac{\pi h}{6} \begin{bmatrix} 4r_1 + h & 2r_1 + h \\ 2r_1 + h & 4r_1 + 3h \end{bmatrix} \begin{Bmatrix} q_1 \\ q_2 \end{Bmatrix}. \tag{B.5}$$

Only the inner radius and the radial length of the element edge are needed. The matrix is the function `get_load_matrix` in `fea_core.py`.

B.3. Uniform traction

For a uniform traction, $q_1 = q_2 = q$, and Equation (B.5) reduces to

$$\{Q\} = \frac{2\pi q\, h}{6} \begin{Bmatrix} 2r_1 + r_2 \\ r_1 + 2r_2 \end{Bmatrix}. \tag{B.6}$$

The outer node receives more load than the inner node, because the outer half of the edge sweeps a larger ring of area. This is the main difference from a plane element, where each node receives $qh/2$. Two checks follow:

- The two loads sum to $\pi q(r_2^2 - r_1^2)$, the traction times the area of the annulus swept by the edge.
- For the element on the axis ($r_1 = 0$, $r_2 = h$), $\{Q\} = \frac{1}{3}\pi q h^2 \{1\ \ 2\}^T$. The axis node carries only a third of the load on the first ring.

When the edge loads are assembled, each interior face node collects a contribution from both edges it belongs to. The total over the face is $\pi a^2 q$. Because the loads depend only on the node positions along $z = 0$, the derivation also holds for distorted meshes, as long as the loaded face stays flat. The nodal-loads test (Section 3.9.2) confirms the assembled loads against this result.

C

Stability of the Central Difference Method

This appendix shows why the central difference method is stable only for $\omega_{max}\Delta t < 2$, which gives the critical time step of Equation (3.18). The argument follows Belytschko et al. (2000, ch. 6) and Hughes (1987, ch. 9).

C.1. Reduction to single-mode equations

The undamped semi-discrete equations of motion are $[M]\{\ddot{d}\} + [K]\{d\} = \{F\}$. The natural frequencies ω_i and mode shapes $\{\phi_i\}$ satisfy $[K]\{\phi_i\} = \omega_i^2[M]\{\phi_i\}$. With the mode shapes normalised so that $\{\phi_i\}^T[M]\{\phi_j\} = \delta_{ij}$, the displacement can be written as $\{d\} = \sum_i \{\phi_i\} y_i$.

The central difference update is linear and uses constant matrices, so substituting this expansion and premultiplying by $\{\phi_i\}^T$ separates it into one independent equation per mode:

$$y_{n+1} - 2y_n + y_{n-1} + \omega_i^2 \Delta t^2 \, y_n = \Delta t^2 f_n, \qquad f_n = \{\phi_i\}^T \{F\}_n. \tag{C.1}$$

The full system is stable if, and only if, every one of these single-mode equations is stable.

C.2. Single-mode stability

The forcing term does not affect stability, so consider the free response. Substituting $y_n = \mu^n$ into Equation (C.1) with $f_n = 0$ gives the characteristic equation

$$\mu^2 - (2 - \Omega^2)\mu + 1 = 0, \qquad \Omega = \omega_i \Delta t. \tag{C.2}$$

The product of the two roots is $\mu_1 \mu_2 = 1$. There are three cases:

- $\Omega < 2$: then $|2 - \Omega^2| < 2$, the roots are a complex conjugate pair, and both have $|\mu| = 1$. The response oscillates with constant amplitude, as the exact solution does.
- $\Omega = 2$: there is a double root $\mu = -1$ and the response grows linearly in time.

- $\Omega > 2$: the roots are real and, since their product is one, one of them has $|\mu| > 1$. The response grows exponentially from any round-off error.

Stability therefore requires $\omega_i \Delta t < 2$ for every mode. The highest mode is the most restrictive, so

$$\Delta t < \Delta t_{cr} = \frac{2}{\omega_{max}}. \tag{C.3}$$

C.3. Notes for this solver

Why an eigenvalue rather than a CFL estimate. A Courant (CFL) estimate, $\Delta t \approx L_{min}/c$, is exact for a 1D bar with lumped mass but is not guaranteed to be conservative in general. In 2D the highest frequency also depends on Poisson's ratio, the element aspect ratio and the mass matrix. The solver therefore finds ω_{max} directly from the eigenvalue problem on the constrained system, with the axis u_r DOFs removed, and uses $\Delta t = 0.9\,\Delta t_{cr}$.

Effect of the mass matrix. A consistent mass matrix gives a higher ω_{max} than a lumped one, so it needs a smaller time step. For a uniform 1D bar with element length L_e, $\Delta t_{cr} = L_e/c_0$ for lumped mass, $L_e/(\sqrt{3}\,c_0)$ for consistent mass, and $\sqrt{2/3}\,L_e/c_0$ for the combined matrix with $\beta = 0.5$. This is why Δt_{cr} is recomputed for each mass matrix.

Period error. For $\Omega < 2$ the numerical frequency $\bar{\omega}$ of each mode satisfies $\sin(\bar{\omega}\Delta t/2) = \omega \Delta t/2$, so $\bar{\omega} \geq \omega$: the time integration makes waves travel slightly too fast, and the effect grows as Δt approaches Δt_{cr} (Hughes, 1987, ch. 9). The lumped mass matrix has the opposite spatial error, so the two partly cancel when lumped mass is run close to Δt_{cr}. With consistent mass the two errors have the same sign and add.

D

Discrete Energy and Momentum Balance

This appendix shows why the energy balance in Section 3.9.3 closes to round-off when half-step velocities are used, but only approximately with on-step velocities. It also shows why axial momentum is conserved exactly.

D.1. Half-step form of the central difference method

Define the half-step velocity $\{v\}_{n+1/2} = (\{d\}_{n+1} - \{d\}_n)/\Delta t$. The central difference update can then be written as

$$[M](\{v\}_{n+1/2} - \{v\}_{n-1/2}) = \Delta t(\{F\}_n - [K]\{d\}_n). \tag{D.1}$$

The on-step velocity used for T in the report is the average of the two half-step velocities, $\{v\}_n = (\{d\}_{n+1} - \{d\}_{n-1})/(2\Delta t) = \frac{1}{2}(\{v\}_{n+1/2} + \{v\}_{n-1/2})$.

D.2. Energy

Define a half-step kinetic term and a half-step stiffness term,

$$\hat{T}_{n+1/2} = \tfrac{1}{2}\{v\}^T_{n+1/2}[M]\{v\}_{n+1/2}, \qquad \hat{U}_{n+1/2} = \tfrac{1}{2}\{d\}^T_n[K]\{d\}_{n+1}. \tag{D.2}$$

Premultiplying Equation (D.1) by $\{v\}^T_n$ and using the symmetry of $[M]$ and $[K]$, each side becomes a difference between successive half steps:

$$\{v\}^T_n[M](\{v\}_{n+1/2} - \{v\}_{n-1/2}) = \hat{T}_{n+1/2} - \hat{T}_{n-1/2}, \tag{D.3}$$

$$\Delta t\,\{v\}^T_n[K]\{d\}_n = \hat{U}_{n+1/2} - \hat{U}_{n-1/2}. \tag{D.4}$$

With $\hat{E}_{n+1/2} = \hat{T}_{n+1/2} + \hat{U}_{n+1/2}$,

$$\hat{E}_{n+1/2} = \tfrac{1}{2}\{v\}^T_{n+1/2}[M]\{v\}_{n+1/2} + \tfrac{1}{2}\{d\}^T_n[K]\{d\}_{n+1}, \tag{D.5}$$

Equation (D.1) becomes $\hat{E}_{n+1/2} - \hat{E}_{n-1/2} = \Delta t \, \{F\}_n^T \{v\}_n$, and summing over all steps gives

$$\hat{E}_{n+1/2} = \hat{E}_{-1/2} + \sum_{k=0}^{n} \Delta t \, \{F\}_k^T \{v\}_k. \tag{D.6}$$

This is an exact algebraic identity of the scheme, not an approximation, so in floating point it holds to round-off. The bar starts at rest with $\{d\}_0 = \{0\}$, and the pulses start at zero, so $\{F\}_0 = \{0\}$. The bootstrap then gives $\{d\}_{-1} = \{0\}$ and $\hat{E}_{-1/2} = 0$. The sum on the right is the external work W evaluated with on-step velocities and the rectangle rule, which is how the solver accumulates it.

Relation to $T+U$. Writing $\{\bar{d}\} = \frac{1}{2}(\{d\}_n + \{d\}_{n+1})$, the conserved quantity can be rearranged as

$$\hat{E}_{n+1/2} = \tfrac{1}{2}\{v\}_{n+1/2}^T \Big([M] - \tfrac{\Delta t^2}{4}[K]\Big)\{v\}_{n+1/2} + \tfrac{1}{2}\{\bar{d}\}^T [K] \{\bar{d}\}. \tag{D.7}$$

It differs from the usual $T + U$ by terms of order Δt^2. These terms oscillate with the motion rather than accumulate, which is why the on-step $T + U$ shows a small, bounded error instead of a drift.

Equation (D.7) also links energy to stability: $\hat{E}$ is positive for every motion only if $[M] - \frac{\Delta t^2}{4}[K]$ is positive definite, which is the condition $\Delta t < 2/\omega_{max}$ of Appendix C.

D.3. Axial momentum

The total axial momentum is $P = \{e_z\}^T [M] \{v\}$, where $\{e_z\}$ has a one in every axial DOF and a zero in every radial DOF. A rigid axial translation produces no strain, so $[K]\{e_z\} = \{0\}$ and, by symmetry, $\{e_z\}^T [K] = \{0\}^T$. Premultiplying Equation (D.1) by $\{e_z\}^T$ removes the stiffness term:

$$P_{n+1/2} - P_{n-1/2} = \Delta t \, \{e_z\}^T \{F\}_n \quad \Rightarrow \quad P_{n+1/2} = P_{-1/2} + \sum_{k=0}^{n} \Delta t \, \{e_z\}^T \{F\}_k. \tag{D.8}$$

The momentum at each half step therefore equals the impulse applied up to that step, again exactly, with $P_{-1/2} = 0$ because $\{F\}_0 = \{0\}$. This holds for all three mass matrices. It relies on the axial motion being unrestrained: the only essential boundary condition, $u_r = 0$ on the axis, does not act on any axial DOF.

E

Mass Matrices

This appendix derives the three mass matrices summarised in Section 3.3, following Cook et al. (2001, ch. 11). The implementation is the function `build_mass_matrices` in `fea_core.py`.

E.1. Consistent Mass Matrix

The consistent mass matrix is derived from the kinetic energy of an element,

$$T = \frac{1}{2}\int_V \rho\,\{\dot{u}\}^T\{\dot{u}\}\,dV. \tag{E.1}$$

The velocity field is interpolated with the same shape functions as the displacement field:

$$\{u\} = [N]\{d\} \quad \therefore \quad \{\dot{u}\} = [N]\{\dot{d}\}. \tag{E.2}$$

Substituting Equation (E.2) into Equation (E.1), and noting that the nodal velocities do not vary over the element, gives

$$T = \frac{1}{2}\int_V \rho\,\left([N]\{\dot{d}\}\right)^T\left([N]\{\dot{d}\}\right)dV = \frac{1}{2}\{\dot{d}\}^T\left(\int_V \rho\,[N]^T[N]\,dV\right)\{\dot{d}\} = \frac{1}{2}\{\dot{d}\}^T[m_e]\{\dot{d}\}, \tag{E.3}$$

where the term in brackets is the element mass matrix,

$$[m_e] = \int_V \rho\,[N]^T[N]\,dV. \tag{E.4}$$

It is called consistent because it uses the same shape functions as the stiffness matrix. Each axisymmetric element represents a ring of material, so $dV = 2\pi r\,dr\,dz$, and mapping to natural coordinates with $dr\,dz = |J|\,d\xi\,d\eta$ gives

$$[m_e] = \int_{-1}^{1}\int_{-1}^{1} \rho\,[N]^T[N]\,2\pi r\,|J|\,d\xi\,d\eta \approx \sum_{i=1}^{3}\sum_{j=1}^{3} \rho\,[N]^T[N]\,2\pi r\,|J|\,w_i w_j. \tag{E.5}$$

The integrand is a polynomial of at most degree four in each natural coordinate, so the 3×3 rule used for the stiffness matrix integrates it exactly (Appendix A). The element matrices are

assembled into $[M_{cons}]$ with the same degree-of-freedom mapping as $[K]$. Because $[m_e]$ couples the inertia of neighbouring nodes, $[M_{cons}]$ is banded and its inverse is dense.

E.2. HRZ Lumped Mass Matrix

Lumping replaces the consistent matrix with a diagonal one, whose inverse is simply the reciprocal of each diagonal term, so each degree of freedom can be updated independently. HRZ (Hinton–Rock–Zienkiewicz) lumping keeps only the diagonal terms of $[m_e]$, which already reflect where the mass lies, and scales them so that the element mass is conserved in each direction.

Let m_i^r and m_i^z denote the diagonal terms of $[m_e]$ for the radial and axial degrees of freedom of node i. The total mass of the element is ρV_e, where the ring volume is

$$V_e = \int_{-1}^{1}\int_{-1}^{1} 2\pi r\,|J|\,d\xi\,d\eta. \tag{E.6}$$

Each direction is scaled so that its diagonal terms sum to the full element mass,

$$\alpha_r = \frac{\rho V_e}{\sum_{i=1}^{4} m_i^r}, \qquad \alpha_z = \frac{\rho V_e}{\sum_{i=1}^{4} m_i^z}, \tag{E.7}$$

giving the lumped nodal masses

$$\bar{m}_i^r = \alpha_r\, m_i^r, \qquad \bar{m}_i^z = \alpha_z\, m_i^z, \tag{E.8}$$

which form the diagonal lumped element mass matrix

$$[m_e]_{lump} = \mathrm{diag}\,(\bar{m}_1^r,\, \bar{m}_1^z,\, \bar{m}_2^r,\, \bar{m}_2^z,\, \bar{m}_3^r,\, \bar{m}_3^z,\, \bar{m}_4^r,\, \bar{m}_4^z)\,. \tag{E.9}$$

Since $[N]$ interpolates u_r and u_z with the same functions, $m_i^r = m_i^z$ and the two scaling factors are equal; they are kept separate here only to show that mass is conserved in both directions. Assembly gives the diagonal global matrix $[M_{lump}]$, whose inverse is found term by term:

$$[M_{lump}]_{ii}^{-1} = \frac{1}{[M_{lump}]_{ii}}. \tag{E.10}$$

E.3. Combined Mass Matrix

In the spatial discretisation, the consistent and lumped matrices produce dispersion errors of opposite sign: consistent mass makes high-frequency waves travel too fast, and lumped mass too slowly. A weighted average of the two allows these errors to partly cancel:

$$[M_{comb}] = \beta\,[M_{cons}] + (1-\beta)\,[M_{lump}], \qquad 0 \le \beta \le 1, \tag{E.11}$$

with $\beta = 0.5$ used here, so that

$$[M_{comb}] = \frac{1}{2}\,([M_{cons}] + [M_{lump}])\,. \tag{E.12}$$

Like the consistent matrix, the combined matrix is not diagonal, so it does not share the low cost per time step of lumped mass. Its effect on the critical time step is discussed in Appendix C.

F

Weekly Progress Reports

This appendix contains the weekly reports on activities, progress and reflection (WRAPR) submitted during the project, in the order they were written, together with the updated project Gantt chart, which was prepared with the Week 7 report. They are reproduced as submitted.

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Weekly Research and Progress Report (WRAPR)

Project: 019 - Explicit Methods for Stress Wave Propagation
Supervisor: Dr Trevor Cloete
Student: Tim Martin MRTTIM005
Reporting period: 3/08 (Meeting 2) → 6/08 (Meeting 3)
GA(s) demonstrated: GA4 (Investigations, experiments and data analysis) - worked through a truss finite element example in Excel to build a foundational understanding of FEA.
GA9 (Independent learning ability) - Consolidating material learned during the meeting and applied it to example problem.

To-Do List (from Meeting 2, 3/08)

1. [Complete Assignment 1] - done
2. [Complete Assignment 2] – 90% done

[Monday, 3/08]

Group meeting at 8am. Shown how to solve simple truss problem.

Worked on Assignment 1 and completed it after the lecture

[Tuesday, 4/08]

Worked on Assignment 2 and completed the pick and place part.

[Wednesday, 5/08]

Continued with Assignment 2 and got the inverted processed global stiffness matrix. Still left to do is get the modified force vector and then do matrix multiplication to get the nodal displacements

[Thursday, 6/08]

3rd group meeting at 9am. Discussed problems and shortcomings of Assignments 1 and 2. Went through derivation for triangle elements. Worked on Assignment 2 after the lecture. Is still yet to be finished

Planning note

Assignment 2 and 3 will be aimed to be finished by next meeting on the 13/08

Questions / items for the 13/08 meeting

Feedback on notes. What I struggled to grasp quickly/where the notes could be more clear or changed.

Acquire Timoshenko and Goodyear textbook on elasticity

Go through linear algebra proofs to understand matrix manipulation steps

Read through section on minimum strain energy in FEA textbook

Weekly Research and Progress Report (WRAPR)

Project: 019 - Explicit Methods for Stress Wave Propagation
Supervisor: Dr Trevor Cloete
Student: Tim Martin MRTTIM005
Reporting period: 6/08 → 13/08
GA(s) demonstrated: GA4 (Investigations, experiments and data analysis) - worked through a constant strain triangular element structure to build a foundational understanding of FEA.
GA9 (Independent learning ability) - Consolidating material learned during the meeting and applied it to example problem.

To-Do List (from Meeting 2, 6/08)

1. Complete Assignment 4
2. How to get all strain energy properties (Timoshenko Theory of Elasticity Textbook)
3. Understand minimum potential energy method from Cook FEA textbook
4. Understand Shape Functions and why they are used

[Thurs, 6/08]

Group meeting at 9am. Lecture on theory behind CST method.

Shape Function Video: https://www.youtube.com/watch?v=zf9ucVTlGeQ

[Fri, 7/08]

Went through derivation of plane stress elasticity matrix. Started implementing inputs in Python script and created Plane stress matrix.

[Sat, 8/08]

Got ESM and struggled with getting ESM to GSM without hardcoding the placement of the ESM. Created an indexing array that tracks the location that the ESM should add to in the GSM. Looped through rows and columns of GSM and efficiently placed ESM to generate correct GSM. Worked on getting processed vector completed. Finally got deflections for each DOF and plotted graph overlayed to original structure.

[Sun, 9/08]

No work undertaken.

[Mon, 10/08]

Generating new examples with known solutions to test code. Errors array sizes causing issues when structure setup changes. Troubleshooting code to be able to work with any CST example

[Tues, 11/08]

Finished troubleshooting code and ran example with known solution to test code. Code fully operational

[Wed, 12/08]

Full day of lectures. Group call in evening to help others who weren't finished

[Thursday, 13/08]

Looked into theory behind bilinear quadrilateral element method; which will make up next week's work. 3rd group meeting at 9am-10:10am.

Actionable

Become familiar with theory behind BLQ. Read notes on natural coordinates and mapping as well as Gaussian quadrature. Start assignment 5 if theory completed prior to Monday's meeting (17/08)

Weekly Research and Progress Report (WRAPR)

Project: 019 - Explicit Methods for Stress Wave Propagation
Supervisor: Dr Trevor Cloete
Student: Tim Martin (MRTTIM005)
Reporting period: 13/08 → 20/08
GA(s) demonstrated: GA4 (Investigations, experiments and data analysis) - worked through a bilinear quadrilateral element structure to build a foundational understanding of FEA. Turned the BLQ code into axial symmetric code.
GA9 (Independent learning ability) - Consolidating material learned during the meeting and applied it to example problems. Researched axial symmetry and how to setup FEA code with axial symmetry.

To-Do List (from Meeting 2, 17/08)

1. Complete Assignment 5
2. Look through axial symmetry and figure out what each matrix and vector should be and how it differs from 2D BLQ
3. Can axial symmetric model, model torsion in element?

[Thurs, 13/08]

Group meeting at 9am with lecture on bilinear quadrilateral element method.

[Fri, 14/08]

Gaussian Quadrature notes and video to help understanding:
https://www.youtube.com/watch?v=Hu6yqs0R7GA

[Sat, 15/08]

Started code for assignment 5(BLQ). Watched video on 2 point Gauss Quadrature:
https://www.youtube.com/watch?v=sefcWt9cd9s.

[Sun, 16/08]

No work undertaken.

[Mon, 17/08]

Group meeting at 8am. Code structure outlined. Functions created: Partial derivative of shape function, gauss point scheme and partial derivative function for jacobian.

[Tues, 18/08]

Completed BLQ Code. resused code from CST script as creating GSM from ESM has same loop structure just with different bounds. troubleshooting with AI, found error in partial function with signs as well as in [E] matrix wtih incorrect v term getting raised to power of 2. ESM was not being reset after each element loop so it just kept adding to itself. All these fixes led me to get same GSM. Code to get from GSM to output identical to CST so copied code over. YT vid on Axial Symmetry code:
https://www.youtube.com/watch?v=8lImesNTLm4 and
https://www.youtube.com/watch?v=P0VYvz9WtD0

[Wed, 19/08]

Researched axial symmetry code and how to go from 2D to 3D. Deep dive into why the derivation is the way it is and how the strain in theta direction doesn't need to be transformed to natural coordinates before going to physical as the theta direction is just a geometric property of the element being revolved around 360 degrees and hence it doesn't have a derivative that affects displacement field in natural coords. Figured out what stayed the same such as the shape function and Gamma and Jacobian matrix. Worked out how matrix B changes and how to get radius that's used in B matrix and final summation formula for ESM. Got AI to give me an example to test my code. GSM and deflections lined up.

[Thursday, 20/08]

Further error checking for axial symmetry. Found that force balances only in z direction and not in r direction. After checking code for errors, none could be found. Found out that forces don't have to balance in the r direction as force balance assumes the structure is at minimum energy, however there was nothing from stopping the ring from stretching in the r direction due to the force applied which is exactly what happens. The deflection matrix returns the correct deflections that account for this stretch, but the force balance still just returns the force that the ring experiences in the r direction.

Actionable

Complete BLQ code and look into axial symmetry and try derive method for axial symmetry. Look into how torsion in axial symmetry can be modelled (24/08)

Weekly Research and Progress Report (WRAPR)

Project: 019 - Explicit Methods for Stress Wave Propagation
Supervisor: Dr Trevor Cloete
Student: Tim Martin (MRTTIM005)
Reporting period: 20/08 → 27/08
GA(s) demonstrated: GA4 (Investigations, experiments and data analysis) – Implemeted stress function that returns stresses at each gauss point. Ran patch test to verify that elements can give exact solution with fine enough mesh.
GA9 (Independent learning ability) – Consolidated theory and learnt theory for new sections (Direct Explicit Integration, Mass Matrices, Stability)

To-Do List (from Meeting 2, 17/08)

- Calculate stress at gauss points
- Do patch test and verify elements
- Derive 3-point Gaussian Quadrature
- Read through past students' reports
- Read through past students' code and understand all functions
- Implement Direct Explicit Integration Scheme (Central difference)

[Fri, 21/08]

Group meeting going over theory behind BLQ and how to transform code into axial symmetric BLQ. Meeting takeaway: I don't understand the theory as well as I thought behind method. Tasks given for the week. Am slightly ahead of schedule, but this is where I want to be so that if anything takes longer there won't be any stress. Can expand scope once completed project objectives. Continued looking through theory on Axial Symmetric BLQ

[Fri, 21/08]

Fixed up code. Made neater by creating more functions to reduce lines of code in main code section.

[Sat, 2/08]

No work undertaken.

[Sun, 23/08]

No work undertaken.

[Mon, 24/08]

Group Meeting with Dr Cloete. Researched how to find stresses in structure after deflections have been found. 2 methods found namely {Stress} = [E][B]{d} or {Stress} =[E][delta][u]

first formulation is easier to implement as E and B are already calculated and {d} just needs to be readjusted to only contain deflections of the element I'm looking at. Structure of code will be creating new loop that iterates over gauss scheme nested inside gauss point scheme to receive stresses at each gauss point.

[Tues, 25/08]

Setting up 2D BLQ code to return stresses. Struggling with dimensions and indexing storing of stress vector as its shape 3x1. 3D array/matrix created to store it (Shape[Numelements, GaussPointScheme**2, 3]) each element calculates stress at each gauss point so this implies having rows equal to elements. Then number of gauss points is just square of the scheme and since stress in 2D is a tuple which holds Stress in x and z as well as shear in xz.

Since stresses are returned at Gauss Points and not nodes, is interpolating stresses to nodes valid? Is it necessary? Is stresses at gauss points sufficient?

[Wed, 26/08]

Patch test theory and implementation. Tested with textbook example. Validated for sum of forces, constant stress in x for each gauss point and interior node displacement matched same linear field of boundary nodes.

[Thursday, 27/08]

Completed Gaussian Quadrature derivation for 3-point. Cook FEA Textbook explicit direct integration theory understood.

Weekly Research and Progress Report (WRAPR)

Project: 019 - Explicit Methods for Stress Wave Propagation
Supervisor: Dr Trevor Cloete
Student: Tim Martin (MRTTIM005)
Reporting period: 28/08 → 03/08
GA(s) demonstrated: GA4 (Investigations, experiments and data analysis) – Implemented mass matrix and explicit integration scheme to FEA problem in axial symmetric Quad elements.
GA9 (Independent learning ability) – Consolidated theory and learnt theory for new sections (Direct Explicit Integration, Mass Matrices, Stability)

To-Do List (from Meeting 2, 31/08)

- Re derive 3-point gauss quadrature without shortcuts and method
- Implement simplest case for explicit integration scheme for FEA
- Consolidate Gantt Chart and make needed adjustments
- Read through past students' reports
- Read through past students' code and understand all functions

[Fri, 28/08]

Fixed up code. Made neater by creating more functions to reduce lines of code in main code section.

[Sat, 29/08]

No work undertaken.

[Sun, 30/08]

No work undertaken.

[Mon, 31/08]

Last group meeting. consolidated theory from the meeting.

[Tues, 01/09]

Transferred stress code from BLQ to axial symmetric BLQ. Noticed just how much stiffer a ring is against deflection compared to 2D BLQ. this makes sense as 2D had thickness of 1mm and here the ring is 2* pi *r which is around 75 times larger. Completed derivation for equivalent nodal load for 2D BLQ and axial symmetric.

[Wed, 02/09]

Started explicit integration scheme code structure: finding what I need, what i know and what I don't know/need to calculate

[Thursday, 03/09]

Got simplest case code running: Assume starts at rest with no displacement/residual stress. Assume constant force. Mass row sum lumping. Started working on excerpt assignment

WRAPR WEEK 7

Project:	019 – Explicit Methods for Stress Wave Propagation
Supervisor:	Dr Trevor Cloete
Student:	Tim Martin (MRTTIM005)
Reporting period:	7/09/2026 – 14/09/2026

Graduate Attributes Demonstrated

GA4 (Investigations, experiments and data analysis) – verification and debugging of the explicit integration scheme.

GA9 (Independent learning ability) – Using direction gained from meeting to plan trajectory and goals for the rest of the project. Independent research into topics such as Pochhammer–Chree wave propagation, which accounts for different-frequency stress waves travelling at different speeds and the need for the model to reflect this.

GA11 (Project Planning and Management) – Progress report with a new Gantt chart to check past, current, and future project trajectory.

Progress Log

7/9 Integration scheme blows up even with extremely low step size (1×10^{-10}). Wave speed for current properties is 5048 m/s, relating to a wave speed of magnitude order 1×10^{-7}.

8/9 Errors with expected mass when using the diagonalised mass matrix in calculations.

9/9 Fixed assumed simplification of the HRZ lumped mass matrix. T_{crit} found to be 2×10^{-7}, in the ballpark of the hand calculation.

10/9 Fixed bug with masses not being scaled correctly under HRZ lumping – was off by a factor of 2 compared to mass from volume × density versus element mass calculated in the mass matrix creation loop.

11/9 Collecting data at each timestep to allow plotting of useful information.

12/9 Attempting to plot stress vs. time.

13/9 Reading through Mr Mareya's code from the prior year.

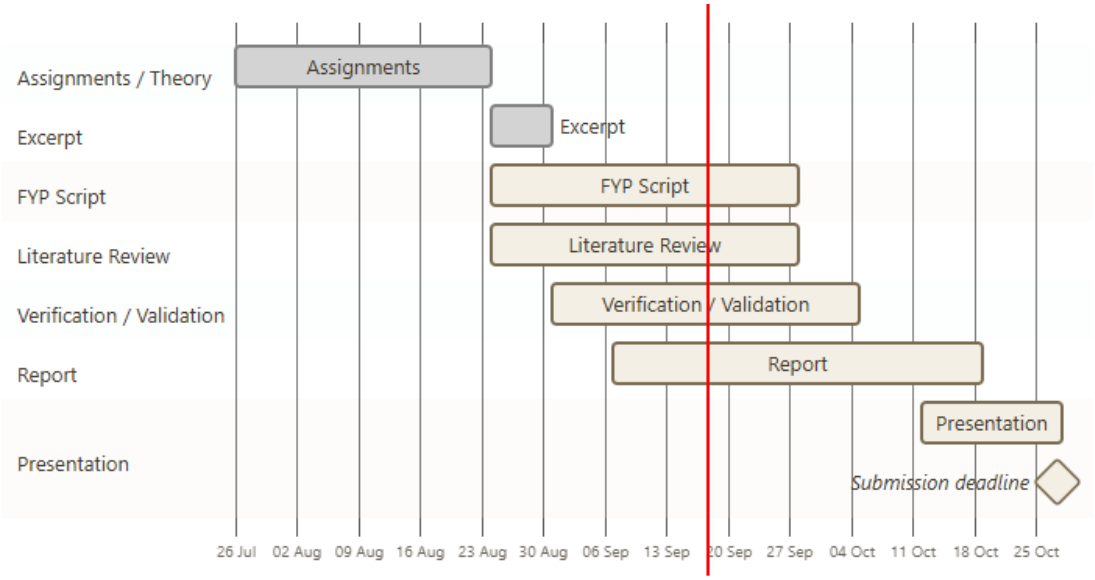
14/9 First individual meeting – direction and goals set for the project. Deriving the general equivalent nodal loading form to be used in the code.

15/9 Completed code that sets up input arrays/vectors and force vectors, along with creating a uniform bar mesh from user input. Generalised the code to bars only, as Hopkinson bars are the project scope.

16/9 Completed strain vs. time plot comparing consistent, lumped, and combined mass matrices, using the input GWS forcing function.

17/9 Confirmed findings from Reuben's MSc research paper that axial mesh density is more important than radial density for convergence – verified with a mesh study that converges using the lumped, consistent, and combined mass matrix assembly methods. Same finding proven from a different diagnostic angle than Reuben et al.'s paper.

Project Progress Gantt Chart



On track with project. Have met scope of project currently, but need more verification and validation. This gives some freedom regarding extending scope to incorporate additional functionality of code. Pochammer-Chree stress theory will be attempted to be verified with my axisymmetric FEA code.

WRAPR WEEK 9

Project: 019 – Explicit Methods for Stress Wave Propagation

Supervisor: Dr Trevor Cloete

Student: Tim Martin (MRTTIM005)

Reporting period: 21/09/2026 – 30/09/2026

Graduate Attributes Demonstrated

GA4 (Investigations, experiments and data analysis) – verification and validation of model including: kinetic and strain energy balance, momentum and impulse balance, mass matrix convergence with mesh refinement using peak von Mises stress at location. Equivalent nodal loading verification and single frequency wave verification

GA9 (Independent learning ability) – Using trajectory from meeting to implement verification scripts for next meeting

GA11 (Project Planning and Management) – Keeping on track with gantt chart and spending week and a half focusing on verification/validation and getting plots ready for report writeup.

Progress Log

21/9 Meeting with progress update and tasks given. Start verification process of code: verifying equivalent nodal loading using constant strain case.

22/9 Continuing with verifying equivalent nodal loading. Completed constant axial strain equivalent nodal verification script. Started single-frequency (monochromatic) wave verification test. I let a single pure frequency (as close to it as possible) pulse propagate through the bar and let it reflect. In a non-dispersive regime: wave should travel at constant speed, reflect off the free end and invert, should reflect back looking the same(just reflected) as original stress wave propagated. Whats being verified: reflection boundary condition and numerical dispersion introduced purely by discretization.

23/9 Troubleshooting script. Needed to change wave to a tapered wave instead of a sharp cut-off to get rid of most dispersive effects that distorted wave. Instead of manually entering the time I want to observe, I observe close to the reflection boundary and invert and overlay strain-time plot over each other from before reflection with after reflection. Plot matches well, however the peaks are off. Need to quantify accuracy. Used RMS values and refined mesh to prove convergence and verify the model's physics. using Envelope comparison instead of raw waveform comparison. Used the Hilbert transform to extract the amplitude envelope of both original and reflected snapshots, and compared those. This strips away the peak-to-peak jitter and shows whether the overall shape survived. Got RMS from over 11% to 5% by refining mesh alongside Hilbert envelope plots.

24/9 Started convergence study script for different mass matrix formulations.

25/9 completed convergence study where peak von Mises stress at a chosen location is compared to each mass matrix formulation and over mesh refinements

26/9 checking mass matrix convergence with small sample size. currently appears to diverge, but mesh not refined enough to be certain. Started looking into kinetic and strain energy balance/fractions

27/9 no work undertaken

28/9 Continued code on energy balance and plotting of comparisons for the different mass matrix formulations.

29/9 Finished energy balance validation script. Machine epsilon error gets to maximum of 30. since gap between smallest representable number is 2.22e-16, floating relative error can get to 1.1e-16. since rounding errors accumulate from massive matrix and vector multiplication, the highest machine error is 30 $\times$ 2.2e-16, so about 6.7e-15. This proves energy isn't unexpectedly lost or gained. Started researching impulse and momentum balance.

30/9 Completed momentum and impulse check. Verified with impulse and momentum agreeing and relative error of 10e-14. Physical impulse values also line up with applied force. Looking into Pochammer-Chree theory

WRAPR WEEK 11

Project: 019 – Explicit Methods for Stress Wave Propagation

Supervisor: Dr Trevor Cloete

Student: Tim Martin (MRTTIM005)

Reporting period: 01/10/2026 –14/09/2026

Graduate Attributes Demonstrated

GA4 (Investigations, experiments and data analysis) – validating model to Pochammer-Chree theory which includes the vibrational modes analysis as well as strain-time comparison of my model against Pochammer-Chree mode 1

GA9 (Independent learning ability) – Using trajectory from meeting to implement validation scripts for next meeting

GA11 (Project Planning and Management) – Keeping on track with gantt chart by finishing up coding and getting into planning and writing of report

Progress Log

01/10 Continued reading into Pochhammer-Chree theory (Graff, Achenbach and Govender et al.) to understand the frequency equation for longitudinal waves in an infinite circular bar. Tidied up the code repository: sorted the verification scripts into their own folder and wrote a README.

02/10 Set up the Pochhammer-Chree frequency equation in Python using SciPy's Bessel functions. Non-dimensionalised frequency and wavenumber so results could be compared directly with Govender et al.

03/10 Started root-finding on the frequency equation to trace the dispersion curves. Main difficulty was keeping track of which root belongs to which mode as frequency increases, since roots can jump between branches.

04/10 No work undertaken.

05/10 Continued root-finding script. Traced modes 1 and 2 over the frequency range of interest and converted the roots into phase velocity against frequency.

06/10 Completed the Pochhammer-Chree dispersion curves. Phase velocity vs frequency plot for the vibrational modes matches Govender et al. This shows the theory side is implemented correctly before comparing it with the FEA.

07/10 Started mode 1 comparison between FEA and Pochhammer-Chree using two measuring stations along the bar and the FFT phase difference between them. Used lumped mass with 300, 450 and 600 axial elements and 5 and 10 radial elements. Could not match the paper's 100/400 elements per wavelength because the model is limited to about 16k elements. FEA agrees with mode 1 up to about 140 kHz, then deviates sharply above about 150 kHz. Govender et al. also

only report agreement below 150 kHz. Cause of the deviation still unresolved, so parked this script.

08/10 Produced the Pochhammer-Chree mode 1 strain-time prediction at a station along the bar and overlaid the FEA strain-time result on it. Started planning the report.

09/10 Report planning. Went through a prior-year report from the same Hopkinson bar project series as a structural reference. Decided which verification and validation results go into the report and in what order.

10/10 Started writing the report, beginning with the methodology. Reworked an earlier methodology excerpt (explicit scheme justification, conditional stability, history independence). It was too wordy and included claims that were never tested.

11/10 No work undertaken.

12/10 Restructured the methodology: each section gives the theory in a few lines, puts the derivation in an appendix, explains it, and then introduces the matching verification or validation test. Wrote the chapter introduction and the Model and Assumptions section (geometry, boundary conditions, material properties and units).

13/10 Started and completed first iteration of the Introduction: background, problem statement and aims.

14/10 Continued with setting up methodology section that shows the theory behind code implemented. Aiming to convey it with explanation on what each part/matrix/vector is/does and also justify it. This gives reader/marker with insight into theory without me just regurgitating the equations.

Appendix

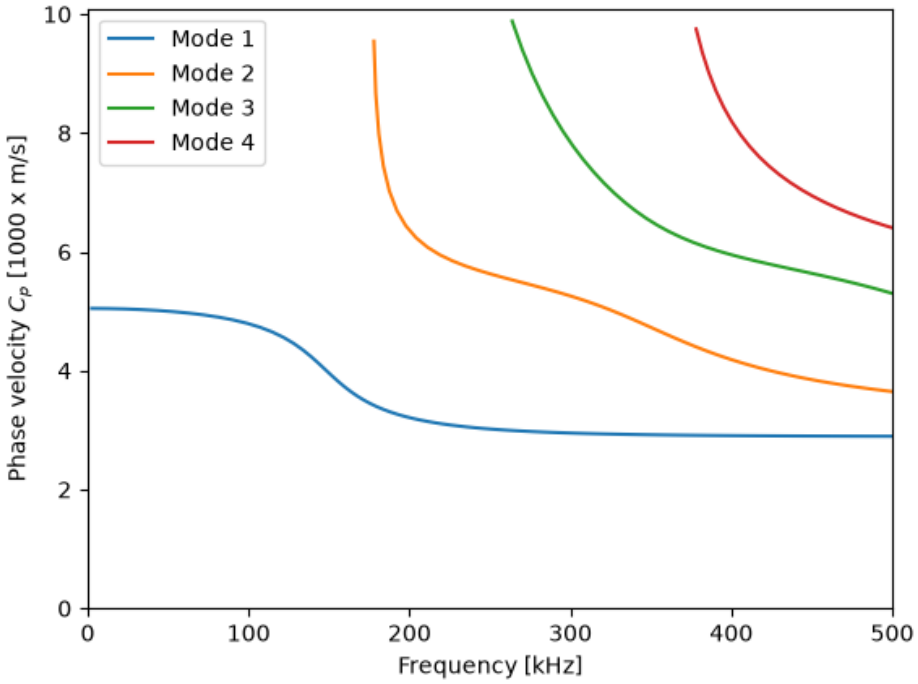


Figure 1: Pochammer-Chree Vibration modes

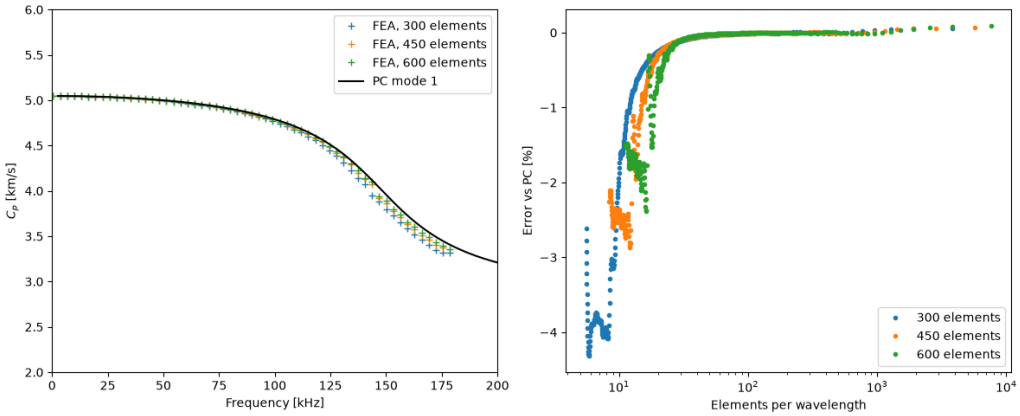


Figure 2: PC phase velocity vs frequency

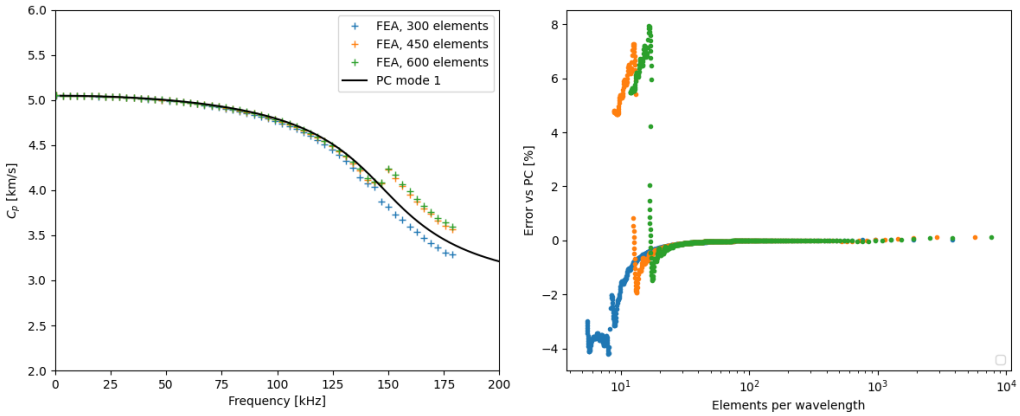


Figure 3: PC phase velocity vs f with finer radial mesh (10 instead of 5 elements)

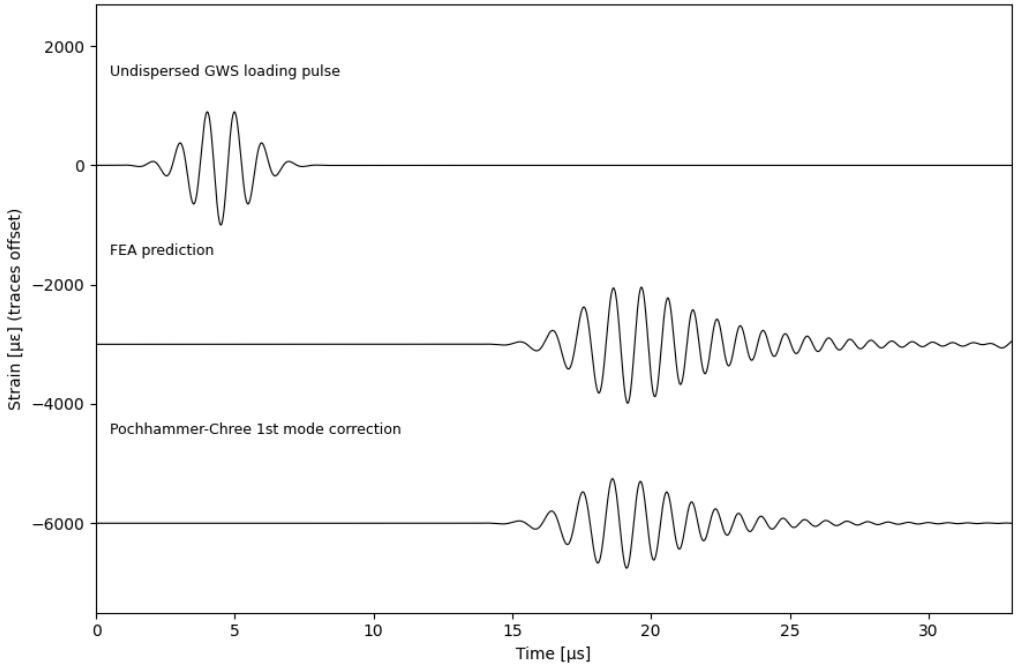


Figure 4: PC strain-time

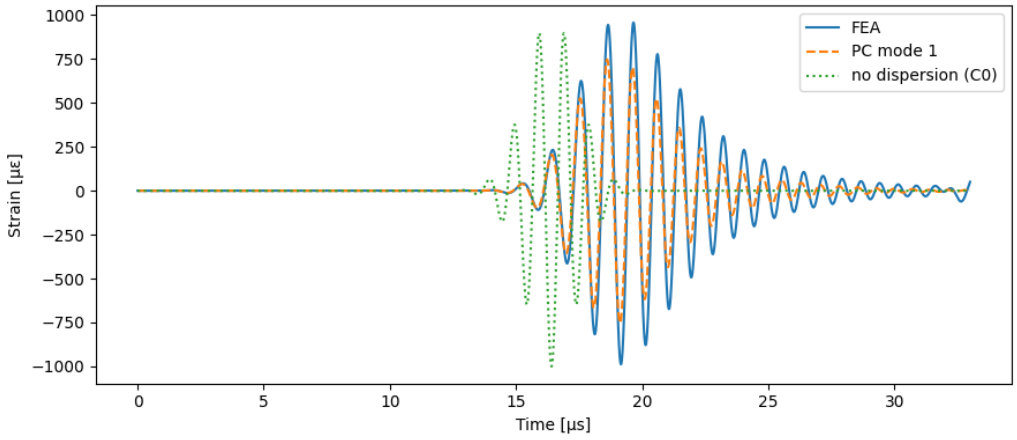


Figure 5: PC strain-time overlayed